

Review

A Scoping Review on Fuzzy Logic Used in Serious Games

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Abstract

This scoping review investigates the use of fuzzy logic in serious games. Articles were searched in nine databases: ACM Digital Library, IEEE Xplore, IOPscience, MDPI, PubMed, ScienceDirect, Springer, Wiley, and Web of Science. The search retrieved 494 articles published between January 2020 and February 2025, of which 28 met the inclusion criteria. Specifically, four research questions were addressed, focusing on the taxonomy of serious games that use fuzzy logic, the characteristics of game design, the purpose and implementation of the fuzzy logic system within the game, and the experiments conducted in the studies. Results reported that 80% of the studies focused on educational serious games, while 20% addressed health applications. Mouse, keyboard, and smartphone touch screen were the most widely used interaction methods. The adventure genre was the most widely implemented in the studies (35.71%). Fuzzy logic was mainly used for adjusting game difficulty, followed by providing tailored feedback in the game. Mamdani inference was the most widely used inference method in the studies. Although 79% of the studies involved human participants in their experiments, 57% did not perform any statistical analysis of their results.

Keywords: fuzzy logic; game; serious game



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1. Introduction

Video games belong to a key and growing industry for entertainment in our society. The aim of these games is to amuse people. Nowadays, technology is more accessible to people; consequently, serious games have been implemented with several purposes (e.g., education, training, healthcare). The term serious game was coined by Prof. Clark Abt in 1970. There are several definitions for serious games. For instance, Zyda [1] defined a serious game as “a mental contest, played with a computer in accordance with specific rules, that uses entertainment to further government or corporate training, education, health, public policy, and strategic communication objectives”. According to [2], the goal of serious games is “to engage players in a way that is both enjoyable and effective in achieving the intended learning or behavior change outcomes”. Several studies have highlighted their importance. For instance, studies [3,4] have shown that educational serious games play a key role in increasing student engagement, motivation, cognitive abilities, and learning strategies. Additionally, another study [2] showed that serious games focusing on health could encourage physical activity, rehabilitation, healthy eating, quitting smoking, mental health improvement, and reduction of depression and anxiety.

On the other hand, Zadeh [5] defined fuzzy logic as “an attempt at formalization/mechanization of two human capabilities: the capability to converse, reason, and make rational decisions in an [...] environment of imperfect information; and the capability

to perform a wide variety of physical and mental tasks without any measurements and any computations”. Based on [6], fuzzy logic describes fuzziness, calibrates vagueness, and allows degrees of membership so that natural language terms can be applied to represent and manipulate fuzzy terms. According to [7,8], a fuzzy inference system consists of four elements: (i) fuzzification, which transforms crisp input values into linguistic variables; (ii) fuzzy rule base, composed of IF (conditions)–THEN (consequences) rules that map fuzzy inputs to fuzzy outputs; (iii) inference engine, which evaluates the rules to compute the overall fuzzy output; and (iv) defuzzification, which converts the fuzzy output into a crisp numerical value.

Several studies have highlighted the importance of fuzzy logic due to its ability to (i) address real-world problems involving uncertainty, ambiguous and incomplete information, providing reliable results [8,9]; and (ii) compute predictions and offer semantic explanations for the reasoning process and the output [10]. Recently, reviews have been published on the use of fuzzy logic in various applications: aerial vehicles and motor drives [11,12], analysis of human movement [13], hydrology [14], manipulator robots [15], opinion mining [16], and sentiment analysis [17]. Nevertheless, none of these has focused on the use of fuzzy logic in serious games.

Similarly, several scoping reviews have been published regarding serious games on health (e.g., mental health [18–20], motor rehabilitation [21,22], healthcare [23], cancer control [24], treatment of alcohol and drug consumption [25], diabetes mellitus [26]). Furthermore, reviews on the use of artificial intelligence in serious games for healthcare have been published [27,28]; however, these focused only on serious games for health. Additionally, a scoping review by Tolks et al. [28] did not report any studies using fuzzy logic, while a scoping review by Abd-alrazaq et al. [27] reported only six articles using fuzzy logic between 2014 and 2019. Table 1 presents these reviews, including the year of publication, the type of review, the keywords used for the search, and the aim of the review.

Table 1. Summary of reviews on fuzzy logic and reviews on serious games.

Reviews on Fuzzy Logic					
Review	Year	Type	Keywords Used for the Search	Aim	Comment
[11] Q. A. Tarbosh et al. 2020	2020	Literature	---	It focuses on the design, functioning, and impact of rule reduction for fuzzy logic controllers.	No serious games were considered
[12] M. M. Ferdaus, et al. 2020	2020	Literature	---	It presents fuzzy systems related to unmanned aerial vehicles.	No serious games were considered
[13] B. N. Lima et al. 2021	2021	Systematic	Not available. Only abstract was accessible	It explains how fuzzy logic has been used to identify the human movement of healthy people.	No serious games were considered
[14] S. Kambalimath and P. C. Deka 2020	2020	Literature	---	It investigates applications of fuzzy logic in hydrology and water resources.	No serious games were considered

Table 1. Cont.

Reviews on Fuzzy Logic					
Review	Year	Type	Keywords Used for the Search	Aim	Comment
[16] J. Serrano-Guerrero, F. P. Romero, and J. A. Olivas 2021	2021	Literature	“fuzzy logic”, “fuzzy sets”, “pythagorean fuzzy sets”, “type-2 fuzzy sets”, “interval type-2 fuzzy sets”, “neutrosophic fuzzy sets”, “hesitant fuzzy sets”	It focuses on applications of fuzzy logic for opinion mining.	No serious games were considered
[17] S. Vashishtha, V. Gupta, and M. Mittal 2023	2023	Literature	---	It presents how fuzzy logic has been used for sentiment analysis.	No serious games were considered
Reviews on Serious Games					
[18] M. Fitzgerald and G. Ratcliffe 2020	2020	Scoping	“games”, “gaming”, “serious games”	It analyzes studies using serious games and gamification for mental illnesses.	Artificial techniques were not considered.
[19] A. Dewhurst, R. Laugharne, and R. Shankar 2022	2022	Scoping	“serious games”, “videogames”	It studies the feasibility and benefits of serious games in treating mental health disorders.	Artificial techniques were not considered
[20] V. Longley, J. Wilkey, and C. Opdebeeck, 2025	2025	Scoping	“serious games,” “digital games”	It presents outcomes assessed in research on serious games for people suffering from dementia and cognitive impairment	Artificial techniques were not considered
[21] O. Mubin, F. et al. 2022	2022	Scoping	Not available. Only abstract was accessible	It analyzes design features of games created for stroke rehabilitation.	Artificial techniques were not considered
[22] E. Koutsiana, et al. 2020	2020	Scoping	“game,” “serious gaming,” “serious game”	It assesses the role of serious games in upper limb rehabilitation.	Only one article was related to fuzzy logic in the references.
[23] Y. Wang et al. 2022	2022	Scoping	serious gam*“, “video game”, “computer game”	It presents applications of serious games in healthcare.	Artificial techniques were not considered
[24] S. Kim, P. Wilson, and O. Abraham 2024	2024	Scoping	“Video games”, “serious game”, “computer game”	It provides an analysis serious games used for cancer prevention.	Artificial techniques were not considered

Table 1. *Cont.*

Reviews on Serious Games					
[25] J. Martínez-Miranda and I. E. Espinosa-Curiel 2022	2022	Scoping	"serious games", "game", "gamification", "videogame"	It focuses on serious games for the prevention and treatment of alcohol and drug consumption. Moreover, it investigates the use of artificial intelligence techniques in serious games. The authors mentioned that the studies only provided the platform on which the game is played.	Details on artificial intelligence techniques were not found.
[26] E. F. H. Reinders 2024	2024	Scoping	"Gamification", "Game", "serious game", "computer game", "video game"	It presents serious games used for diabetes.	Artificial techniques were not considered
[27] A. Abd-alrazaq et al., 2022	2022	Scoping	Keywords were used for each artificial intelligence technique. Keywords related to this review: "Fuzzy Logic", "Video Games", "serious gam*", "gamification"	It focuses on serious games for healthcare. The authors explored the use of artificial intelligence techniques in this type of serious game.	The authors reported six articles using fuzzy logic for healthcare
[28] D. Tolks, J. J. Schmidt, and S. Kuhn 2024	2024	Scoping	"game", "gamification", "artificial intelligence"	It presents artificial intelligence techniques used in serious games for health.	No articles were found using fuzzy logic as artificial intelligence techniques

Contributions of This Study

This scoping review has the following contributions: (i) maps the state-of-the-art from January 2020 to February 2025, focusing on the use of fuzzy logic in serious games; (ii) explains a taxonomy for serious games that employ fuzzy logic, detailing aspects such as the application area (e.g., education, health), types of player activity (i.e., mental, physical, physiological), feedback modalities to players (e.g., visual, auditory, haptic), interaction methods (e.g., keyboard, mouse, touch screen, body movements), environment (e.g., 2D, 3D, virtual reality), and hardware architecture (e.g., desktop, smartphone, tablet); (iii) identifies game genres, narrative, and gameplay rules used in serious games with fuzzy logic; (iv) describes how fuzzy logic was applied in serious games and how fuzzy inference systems were implemented, including membership functions for fuzzification, inference methods, fuzzy rules, and defuzzification methods; and (v) reports whether human participants were involved in the experiments, what instruments and metrics were used to assess the performance of serious games, and what statistical tests were applied to support the findings of the studies. These contributions will be analyzed based on the research questions.

The rest of the article is organized as follows: Section 2 explains the methodology used to conduct this scoping review, including the identification of research questions, the identification of relevant studies, the study selection criteria, and the data obtained for

each research question. Section 3 presents the results related to the four research questions addressed in this scoping review. Section 4 provides a discussion of the key findings and future directions for each research question. Additionally, the risk of bias of the studies and the limitations of this scoping review are explained. Finally, Section 5 presents the conclusions of this study.

2. Methods

This scoping review was conducted using the framework proposed by [29], which involves five stages: (1) identification of research questions, (2) identification of relevant studies, (3) study selection, (4) charting the data, and (5) reporting the results.

2.1. Identification of Research Questions

This scoping review aims to answer the following research questions (RQ): RQ1—What is the taxonomy of serious games using fuzzy logic (i.e., application area, activity, modality, interaction method, environment, and hardware architecture)?; RQ2—What are the game design characteristics (i.e., game genre, narrative, and game rules—how it is played)?; RQ3—What is the aim of using fuzzy logic in serious games, and how is the fuzzy logic system implemented (i.e., fuzzification, fuzzy inference, defuzzification)?; and RQ4—Were experiments conducted in the studies (i.e., participants, instruments and metrics, key results, and statistical analysis)?

2.2. Identification of Relevant Studies

Two searches were conducted in nine electronic databases: ACM Digital Library, IEEE Xplore, IOPscience, PubMed, MDPI, ScienceDirect, Springer, Wiley, and Web of Science.

The first search used the keywords “fuzzy logic” and “serious game”, while the second search used “fuzzy logic” and “game”. These keywords were selected based on search terms used in the following reviews [16,18–20,22–28]. Electronic databases might provide filters and advanced search options, which were used to refine the results. The search period for all the databases included publications from January 2020 to February 2025. Table 2 summarizes the filters applied in each database for both searches. The first search was conducted between 25 February and 27 February 2025, and the second search was conducted between 2 March and 5 March 2025.

Table 2. Filters and advanced search options applied in the electronic databases.

Database	Filters and Advanced Search Options Used in Electronic Databases
ACM Digital Library	The ACM full-text collection, research articles, and journals
IEEE Xplore	None—in first search Title: game—in second search
IOPscience Web of Science	Open access
MDPI PubMed	None
ScienceDirect Wiley	Open access—in both searches Article title: game—in second search
Springer	Research articles, articles—in both searches Article title: game—in second search

2.3. Study Selection

The following inclusion and exclusion criteria were defined to select the articles that were included in this review.

2.3.1. Inclusion Criteria

Articles included in this review must: (i) use or implement a serious game; (ii) use fuzzy logic in the serious game; (iii) be written in English; and (iv) be published between 2020 and 2025. The search was restricted to studies published between 2020 and 2025 to include and analyze the most recent developments in serious games and fuzzy logic, considering the fast evolution of both fields and their technological applications.

2.3.2. Exclusion Criteria

Articles were excluded if they: (i) were reviews, books, or editorial notes; (ii) had a length of fewer than four pages; (iii) had an editorial note regarding a concern on the manuscript; or (iv) referred to fuzzy logic only theoretically without any computational implementation.

2.4. Chart the Data

Data were extracted from each article to address the research questions as follows.

2.4.1. RQ1—What Is the Taxonomy of Serious Games Using Fuzzy Logic?

To respond to this question specifically, criteria from the taxonomies proposed by [30,31] were used. The following criteria were analyzed for each article:

Application area. This explains the area on which the serious game aims to focus. For example: educational serious games, serious games for health applications, serious games for training, serious games for well-being, etc.

Activity. This involves the activity that the player uses to play the game (i.e., physical exertion, mental, and physiological).

Modality. It refers to the feedback given to the player in terms of the game design (i.e., visual, auditory, haptic, and smell aspects of the game).

Interaction method. This refers to the means used by the player to interact with the game. For example: keyboard, smartphone touch screen, joystick, mouse, brain interface, eye-gaze interface, movement-tracking interface, etc.

Environment. This involves whether the game's graphical interface was implemented in 2D, 3D, or as a virtual environment.

Hardware architecture. This criterion is not considered by [30]; however, it is introduced in [31]. It refers to the platform (arcade, PC, console, smartphone/tablet) on which the game is played.

2.4.2. RQ2—What Are the Game Design Characteristics?

The following aspects were investigated per article to answer this question:

Game genre. This indicator considers the game genres proposed by [32], which are action, adventure, fight, logic, simulation, sport, and strategy.

Narrative. It refers to the story of the game.

Game rules. It explains how the game is played.

2.4.3. RQ3—What Is the Aim of Using Fuzzy Logic in Serious Games and How Is the Fuzzy Logic System Implemented?

This RQ analyzes the following:

Aim of fuzzy logic. This explains the application of fuzzy logic in the serious game.

Fuzzy Logic System. This involves the methods used for fuzzification (i.e., membership functions to convert crisp input values to fuzzy input sets), fuzzy inference (i.e., the rules—If (conditions) Then (consequences) statements—to map the fuzzy input sets to the fuzzy

output sets), and defuzzification (i.e., the method used to convert the fuzzy output to the crisp output value).

2.4.4. RQ4—Were Experiments Conducted in the Studies?

Four parameters were obtained from the articles to answer this RQ:

Participants. It indicates the sample of participants involved in the experiments in the study.
Instruments and metrics. It explains the instruments used to assess the game performance and participants' feedback (e.g., game score, questionnaires, participants' emotions).

Key results. It reports the main findings of the study.

Statistical analysis. It presents the statistical tests that were applied to the results to support the findings.

3. Results

Initially, the searches performed in the nine databases retrieved 494 articles. After deleting duplicated articles, 442 articles were analyzed based on their titles and abstracts. PRISMA protocol [33] was used to include articles (see Figure 1). Articles were excluded in the first screen (i.e., based on their titles and abstracts) because (i) they were reviews, editorial notes, or books (sixty-six articles); (ii) they were not written in English (one article); (iii) they presented editorial concerns (five articles); (iv) they were not accessible (eight articles); and (v) they were out of the scope based on their titles and abstracts (one hundred and eighty-one articles). After reading the 181 articles, 153 articles were excluded because (i) they neither used fuzzy logic nor used or implemented a video game ($n = 5$); (ii) they used a game; however, they did not use fuzzy logic ($n = 105$); (iii) they did not use fuzzy logic ($n = 24$); (iv) the fuzzy logic system was not explained ($n = 1$); and (v) the serious game was not described ($n = 2$). As a result, 44 articles were analyzed in depth; however, 16 articles ([34–48]) were excluded because, even though they used fuzzy logic and a video game, the video game was not a serious game. Consequently, only 28 articles met all the inclusion criteria and are presented in this scoping review.

3.1. Results on RQ1—What Is the Taxonomy of Serious Games Using Fuzzy Logic?

Based on the application area (see Figure 2), most studies implemented or used educational serious games (82.14% of the studies), followed by serious games for healthcare (17.86% of the studies) for (i) hand rehabilitation [49]; (ii) upper limb motor impairment [50]; (iii) cognitive functions in elderly people [51]; (iv) autism in children [52]; and (v) dysgraphia disorder in children [53]. Regarding the educational serious games, they focused on (i) teaching math [54,55]; (ii) teaching programming and computational skills (i.e., HTML [56–58], C++ [59], computational skills for controlling a robot via control commands [60], and support for co-learning between students and robots in classroom environments [61]); (iii) teaching cultural and historical aspects (i.e., citizenship behavior [62], historical events in Indonesia [63,64], Javanese letters [65]); (iv) encouraging consumers to reduce energy consumption [66,67]; (v) teaching electroencephalography electrode placement [68] and electrotherapy techniques [69]; (vi) teaching general knowledge on animals, arts, sports, history, and geography [70]; and (vii) handling emergencies (i.e., learning content on disaster mitigation [71], and simulating pre-evacuation human reactions in fire emergencies [72]).

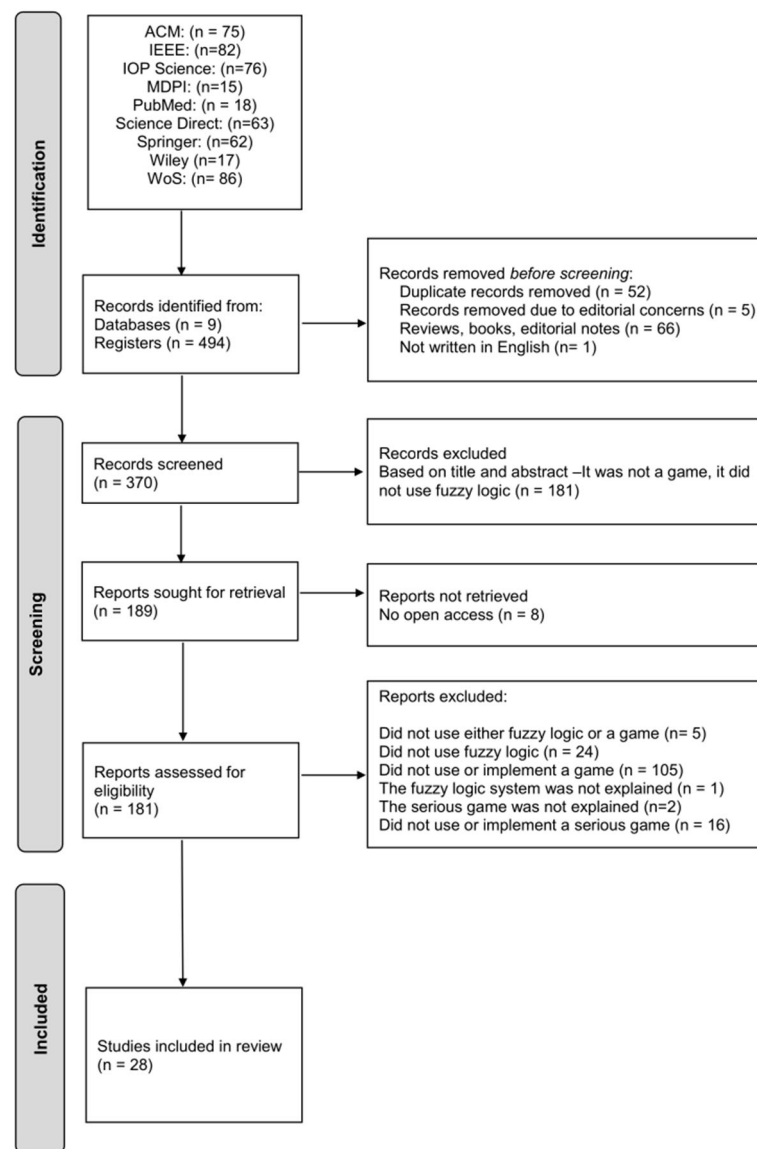


Figure 1. PRISMA flowchart of the searches.

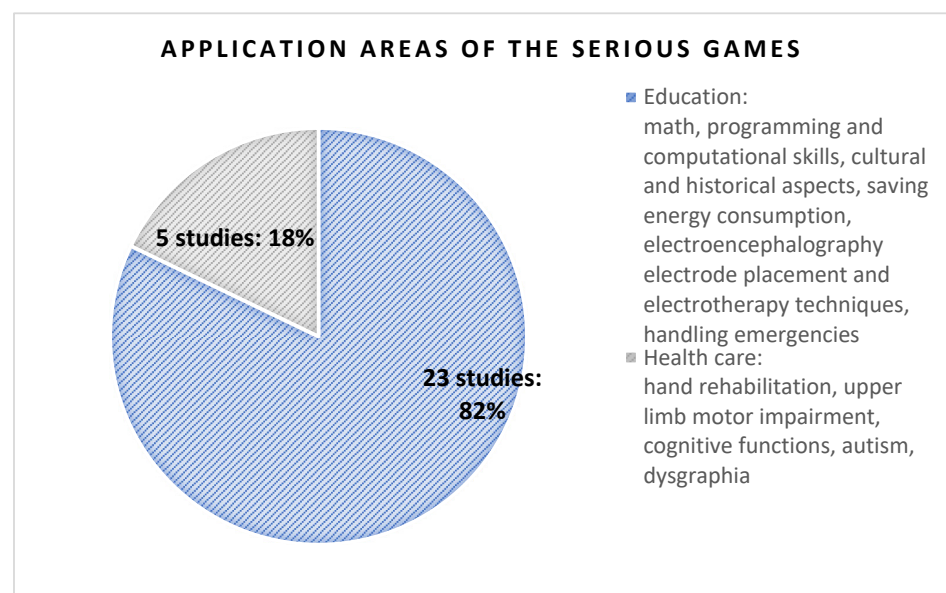


Figure 2. Application areas of the serious games.

In the context of the activity used in serious games (see Figure 3), most studies employed serious games in which the participants needed to perform mental activities (25 studies: 89.28%). Nevertheless, physical exertion during gameplay was required of participants in only four of the studies [49,50,52,53]. Moreover, in only one study [52], the players performed physiological activity to be able to play the game. It is important to note that few studies required the players to perform two activities to play the games (i.e., physical exertion and physiological [52], mental and physical exertion [53]).

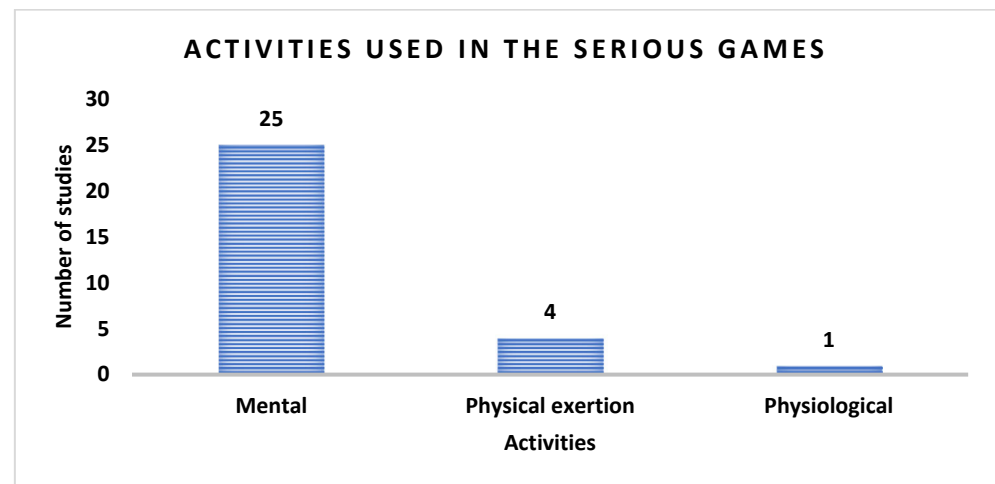


Figure 3. Activities used in the serious games.

Focusing on the modality provided by the games (i.e., the feedback given to the players), only one study did not present the information, and there was insufficient context to infer the modality. The remaining studies used serious games that provided visual feedback. Over one-third of the studies (35.71%: 10 out of 28 studies [51,52,54,55,61,62,70,72–74]) employed serious games that offered visual and auditory feedback to players. It is important to note that none of the studies provided haptic feedback to players.

Regarding the interaction method (see Figure 4), four studies did not indicate it, and the information could not be inferred from the articles. The most widely used interaction methods in the serious games were mouse and keyboard (32%: 9 out of 28 studies [54,55,57,58,60,68,73,75,76]) and the touch screen of a smartphone (32%: 9 out of 28 studies [51,58,62,63,65–67,69,70]).

Other studies have used body movements to play the games. For instance, hand movements detected by a Leap motion controller sensor [49] and by a Kinect depth camera [50,53] were employed to play games for motor rehabilitation and a game for dysgraphia disorder treatment (i.e., a disorder that affects the ability to write in children). Moreover, few studies have integrated virtual reality into their games using head-mounted virtual reality displays as the interaction device [51,72,74].

Few studies [51,54] used the player's voice to interact with the game. Finally, one study [52] employed breathing and acceleration data collected from a smartwatch as the interaction method to help children with autism practice box breathing techniques.

In terms of the environments created in the serious games, it can be seen from Figure 5 that over half of the games were implemented using a 3D environment (61%: 17 out of 28 studies [49–54,56–58,63,65–68,73,75,76]), followed by a 2D environment (32%: 9 out of 28 studies [55,60–64,69–71]). Few studies [72,74] have implemented virtual reality environments in their games.

Focusing on the hardware architectures (i.e., platforms) in which the serious games could be played (see Figure 6), the most widely used platforms were desktops and

laptops (50%: 14 out of 28 studies [49,50,53–55,57,58,60,61,68,72,73,75,76]), followed by smartphones (32%: 9 out of 28 studies [51,58,62,63,65–67,70,74]). Few studies employed tablets [52,69] and virtual reality visors [72,74] to play the serious games. Table 3 summarizes the studies in terms of RQ1.

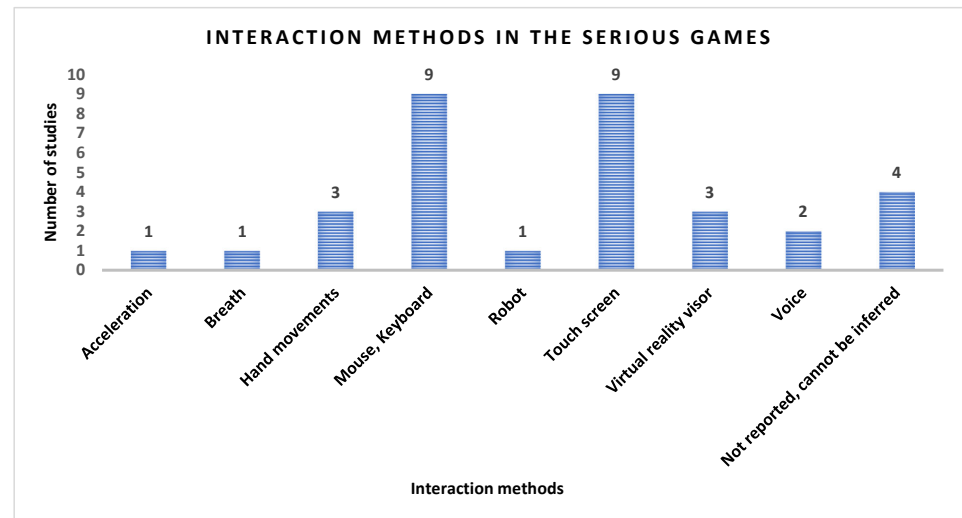


Figure 4. Interaction methods in the serious games.

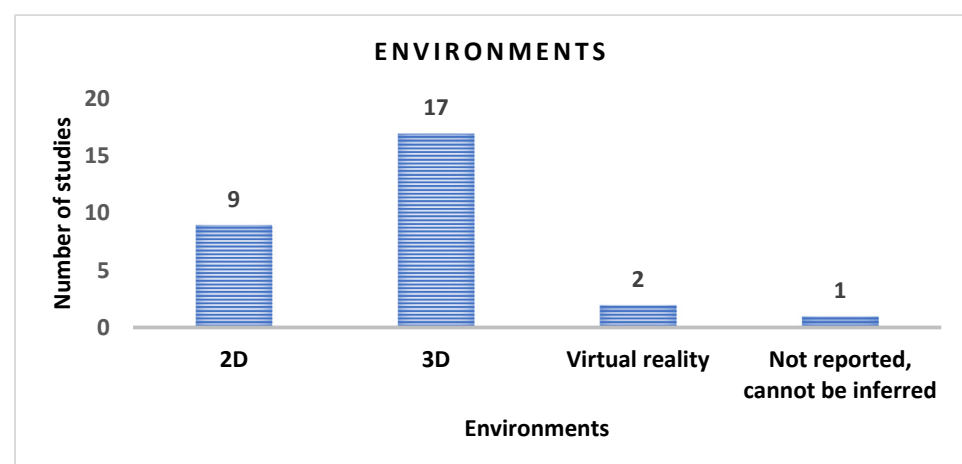


Figure 5. Environments created in the serious games.

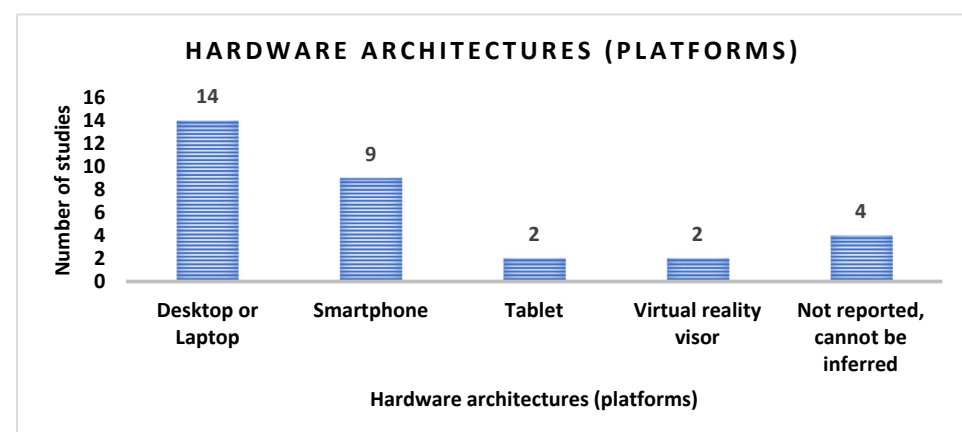


Figure 6. Hardware architecture (platform) on which the serious games could be played.

Table 3. Indicators in the taxonomy of serious games using fuzzy logic.

Reference	Year	Application Area	Activity	Modality	Interaction Method	Environment	Hardware Architecture (Platform)
[60] (Guniš et al., 2025)	2025	Education: to assess computational thinking by controlling a robot through commands (computational skills)	Mental	Visual	Not explicitly indicated Mouse Keyboard (inferred from the article)	Not explicitly indicated 2D (inferred from the article)	Not explicitly indicated Desktop or laptop (inferred from the article)
[49] (García-Ramón et al., 2024)	2024	Healthcare: to assist in hand rehabilitation	Physical exertion	Visual	Hand movements were detected via a Leap motion controller using Ultraleap tracking service version 5.7.2., Control panel version 3.1.0, and Unity package Ultraleap tracking version 6.7.0	Not explicitly indicated 3D: Unity version 2021.3.24f1 (inferred from the article)	Desktop or laptop
[73] (Liu et al., 2024)	2024	Education: to model reactions in virtual characters	Mental	Visual Auditory	Not explicitly indicated Mouse Keyboard (inferred from the article)	Not explicitly indicated 3D (inferred from the article)	Not explicitly indicated Desktop or laptop (inferred from the article)
[75] (Nourian et al., 2024)	2024	Education: to help participants with spatial planning and decision-making in urban redevelopment issues (e.g., residential, commercial, cultural spaces)	Mental	Visual	Not explicitly indicated Mouse Keyboard (inferred from the article)	3D: multi-player online game. The interactive interface was created via the React framework in JavaScript. The maps were included via MapBox. Geospatial information was displayed using Vis.gl. The game engine was implemented via Python 2.7: NumPy for the algebraic processes, Pandas for organizing data, topoGenesis for spatial indexing, and HoneyBee https://github.com/ladybug-tools/honeybee (accessed on 28 September 2025) for solar analyses	Not explicitly indicated Desktop or laptop (inferred from the article)
[62] (Panjaburee et al., 2024)	2024	Education: to encourage digital citizenship behavior	Mental	Visual Auditory	Not explicitly indicated Touch screen. (inferred from the article)	2D	Not explicitly indicated Smartphone (inferred from the article)

Table 3. Cont.

Reference	Year	Application Area	Activity	Modality	Interaction Method	Environment	Hardware Architecture (Platform)
[70] (Tselepati-otis and Alepis, 2024)	2024	Education: to teach general knowledge in terms of animals, arts, sports, history, and geography.	Mental	Visual Auditory	Touch screen	2D: Unity	Smartphone
[56] (Chrysafiadi et al., 2023)	2023	Education: to teach HTML	Mental	Visual	Not indicated and cannot be inferred from the article	3D	Not indicated and cannot be inferred from the article
[74] (Felix et al., 2023)	2023	Education: to improve health professionals’ learning using 360° videos to support women experiencing domestic violence	Mental	Visual Auditory	A smartphone, headphones, and a virtual reality headset	Virtual reality (3D)	Virtual reality headset, smartphone
[50] (Jiang et al., 2023)	2023	Healthcare: to assess the degree of upper limb motor impairment	Physical exertion	Visual	Hand movements were detected using a Kinect depth camera	3D: Unity3D engine and C#	Desktop or laptop
[63] (Jondya et al., 2023)	2023	Education: to teach historical events in Indonesia’s struggle for independence	Mental	Visual	Touch screen Joystick Swiping the screen	2D, 3D	Smartphone
[59] (Krouska et al., 2023)	2023	Education: to teach C++ programming	Mental	Not indicated and cannot be inferred from the article	Not indicated and cannot be inferred from the article	Not indicated and cannot be inferred from the article	Not indicated and cannot be inferred from the article
[66] (Méndez et al., 2023)	2023	Education: to teach and encourage consumers to reduce energy consumption through optimized thermostat usage	Mental	Visual	Not explicitly indicated Touch screen (inferred from the article)	3D	Not explicitly indicated Smartphone (inferred from the article)
[69] (Rueda et al., 2023)	2023	Education: to teach students electrotherapy techniques through simulations of real-world clinical scenarios	Mental	Visual	Touch screen	Not explicitly indicated 2D (inferred from the article)	Two iOS devices: an iPad 2 WLAN + 3G 64 GB of 2011 and an iPad Mini 4 64 GB of 2015.

Table 3. Cont.

Reference	Year	Application Area	Activity	Modality	Interaction Method	Environment	Hardware Architecture (Platform)
[57] (Chrysafiadi et al., 2022)	2022	Education: to teach HTML programming	Mental	Visual	Not explicitly indicated Mouse or keyboard (inferred from the article)	Not explicitly indicated 3D (inferred from the article)	Not explicitly indicated Desktop or laptop (inferred from the article)
[51] (Ghorbani et al., 2022)	2022	Healthcare: to assess cognitive functions in elderly people	Mental	Visual Auditory	Voice, touch screen, and augmented reality	3D: ARCore SDK (Software Development Kit) for Unity Augmented reality	Smartphone Samsung Galaxy S9 Plus smartphone, with a dual 12 MP rear camera to view the augmented reality experience
[71] (Haryanto et al., 2022)	2022	Education: to study key elements of disaster mitigation	Mental	Visual	Not indicated and cannot be inferred from the article	Not explicitly indicated 2D (inferred from the article)	Not indicated and cannot be inferred from the article
[52] (Morales et al., 2022)	2022	Healthcare: to support children with autism in practicing box breathing	Physical exertion Physiological	Visual Auditory	Breathing was measured via a smartwatch Accelerometry data and visual attention measured via an RGB camera	Not explicitly indicated 3D (inferred from the article)	Tablet
[64] (Rachmawati et al., 2022)	2022	Education: to teach historical figures of the Indonesian revolution	Mental	Visual	Not indicated and cannot be inferred from the article	Not explicitly indicated 2D (inferred from the article) Single player	Not indicated and cannot be inferred from the article
[53] (Aziz Hutama et al., 2021)	2021	Healthcare: to assist in the treatment of dysgraphia. Dysgraphia affects the ability to write in children.	Mental Physical exertion	Visual	Hand movements were identified using a Kinect Studio V2 and Visual Gesture Builder. The game and Kinect were connected via Kinect for Windows SDK 2.0	3D: Unity3D and C#	Not explicitly indicated Desktop or laptop (inferred from the article)
[54] (Lara-Alvarez et al., 2021)	2021	Education: to introduce inductive control—i.e., eliciting emotions in the player to improve the learning process—in educational games, specifically in a game designed to teach basic mathematics	Mental	Visual Auditory	Keyboard and voice. The voice was processed using the openEAR toolkit	Not explicitly indicated 3D (inferred from the article)	Not explicitly indicated Desktop or laptop (inferred from the article)

Table 3. Cont.

Reference	Year	Application Area	Activity	Modality	Interaction Method	Environment	Hardware Architecture (Platform)
[65] (Purnamasari et al., 2021)	2021	Education: to learn Javanese letters	Mental	Visual	Touch screen	Not explicitly indicated 3D (inferred from the article)	Smartphone
[76] (Suwindra et al., 2021)	2021	Education: to develop children's characters using cultural wisdom	Mental	Visual	Not explicitly indicated Keyboard (inferred from the article)	Not explicitly indicated 3D (inferred from the article)	Not explicitly indicated Desktop or laptop (inferred from the article)
[68] (Björn et al., 2020)	2020	Education: to teach electroencephalography electrode placement	Mental	Visual	Not explicitly indicated Keyboard, mouse (inferred from the article)	3D	Desktop or laptop
[72] (Bourhim and Cherkaoui, 2020)	2020	Education: to simulate pre-evacuation human reactions in fire emergencies	Mental	Visual Auditory	Head movements and tap controllers were used to navigate in the environment through an HTC Vive head-mounted virtual reality display	Virtual reality (3D): Unity3D 2016 Modeling of the environment was created via AutoCAD and 3Ds Max	Laptop, virtual reality headset
[58] (Chrysafiadi et al., 2020)	2020	Education: to teach HTML programming language	Mental	Visual	Not explicitly indicated Keyboard Mouse Touch screen (inferred from the article)	Not explicitly indicated 3D (inferred from article) Online	Desktop or laptop Smartphone
[61] (Lee et al., 2020)	2020	Education: to assist co-learning between students and robots in classroom environments	Mental	Visual Auditory	Players interacted with the robot and visual programming tools (i.e., Blockly on CodeLab: https://developers.google.com/blockly (accessed on 28 September 2025), or Webduino:Bit) to edit robot behaviors, not to play the Go game interface	Not explicitly indicated 2D (Inferred from the article)	Not explicitly indicated Desktop or laptop (inferred from the article)

Table 3. Cont.

Reference	Year	Application Area	Activity	Modality	Interaction Method	Environment	Hardware Architecture (Platform)
[67] (Ponce et al., 2020)	2020	Education: to teach and encourage consumers to reduce energy consumption through optimized thermostat usage	Mental	Visual	Not explicitly indicated Touch screen (inferred from the article)	Not explicitly indicated 3D (Inferred from the article)	Not explicitly indicated Smartphone (inferred from the article)
[55] (Robles and Quintero M., 2020)	2020	Education: to teach math	Mental	Visual Auditory	Not explicitly indicated Keyboard and mouse (inferred from the article)	2D: online Website: https://www.arcademics.com/ (accessed on 28 September 2025)	Not explicitly indicated Desktop or laptop (inferred from the article)

3.2. Results on RQ2—What Are the Game Design Characteristics?

It can be seen from Figure 7 that most studies used the adventure game genre (35.71%: [52,56–58,62,63,70,73,74,76]), followed by the simulation genre (33.33%: [51,66–69,71,72,75]) in their games. Additionally, action games (14.28%: [49,53–55]) and logic games—specially puzzles—(14.28%: [59,60,64,65]) were implemented in the studies. Finally, only one study [50] created a sport game, and another study [61] created a strategy game.

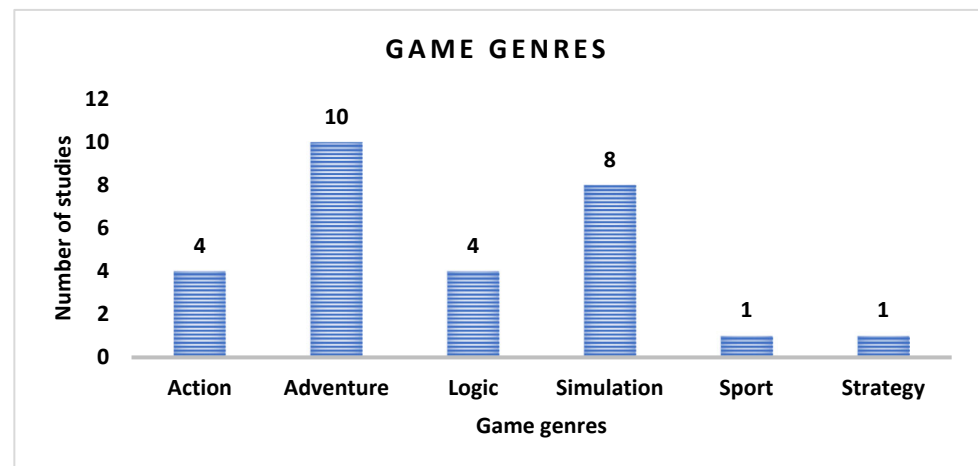


Figure 7. Game genres of the serious games.

Regarding the game narrative, the adventure games involved storytelling elements, such as (i) exploring a water village [73] and an island with castles [56]; (ii) understanding scenarios related to digital citizenship [62] and historical events [63]; (iii) fighting zombies [70]; (iv) following a character through childhood, adolescence, and adulthood in search of healthcare support [74] and figures from Balinese culture [76]; (v) escaping from a locked house [57,58]; and (vi) protecting the world by maintaining the balance of water, wind, earth, and fire [52].

Simulation games focused on scenarios such as (i) urban planning [75], energy consumption [66], and a connected thermostat interface [67]; (ii) clinical situations requiring electrotherapy interventions [69] and the placement of electroencephalography electrodes on a 3D human head [68]; (iii) daily living situations related to cognitive functions [51]; and (iv) disaster mitigation [71] and firefighting scenarios [72].

Logic games included puzzles such as guiding robots to light up boxes [60], solving C++ programming challenges [59], identifying historical figures [64], and working with puzzles on Latin characters [65].

In the context of action games, these involved destroying boxes [49], catching falling apples [53], placing falling objects into the correct containers [54], and running races [55]. Finally, the only sport game was a ping-pong match [50], while the only strategy game required placing black or white stones on a grid [61].

Focusing on how the games were played, most studies (64%: 18 out of 28 studies) required players to select options or objects displayed in the game interface [52,54,55,57–59,62,63,65–69,71,72,74–76]. Some games provided control commands, such as move forward, turn right, turn left [49,54,60], and throw [49] or shoot [70]. Additionally, certain games involved navigating through the game environment [56,63,72], while others required players to mimic striking postures [50] or catching movements [53]. Table 4 explains further details on RQ2.

Table 4. Game design.

Reference	Year	Game Genre	Narrative/Storyline	Game rules How the Game is Played
[60] (Guniš et al., 2025)	2025	Logic—Puzzle	Light-Bot is an educational game that teaches computational thinking. Players create a set of instructions (an algorithm) to guide a robot to light up all the blue boxes in each level.	Players must light up blue boxes using the following commands: move forward, turn 90° right, turn 90° left, light the box, call function 1, call function 2, and jump (either up or down).
[49] (García-Ramón et al., 2024)	2024	Action—shooter	A wall and boxes are to be destroyed by throwing a ball.	The ball is controlled using hand movements: • A fist with the right hand to throw the ball. • Ulnar deviation with the left hand to move to the left in the game scene. • Radial deviation with the left hand to move to the right in the game scene. • Wrist extension with the left hand to move upward in the game scene. • Wrist flexion with the left hand to move downward in the game scene.
[73] (Liu et al., 2024)	2024	Adventure game	The virtual character responds to environmental stimuli based on perception, motivation, and emotion. The virtual character explores a waterside village. Stimuli are displayed to the virtual character (e.g., a barking dog, a fruit stand, a friend, a dancer). Based on perception and motivation, the virtual character: • Escapes from the dog due to the danger. • Eats fruit if hungry. • Greets a friend. • Enjoys watching people dancing. Facial expressions from the virtual character are displayed to show the emotion in each stimulus.	The player watches how virtual characters react to different stimuli and situations in the game.
[75] (Nourian et al., 2024)	2024	Simulation game	Equicity game (multiplayer online game). It simulates participatory urban planning. Players act as stakeholders (e.g., planners, citizens, developers), each with different levels of control, interest, and goals. Players collaborate and negotiate to co-design a spatial configuration of an urban district.	Players make decisions on how to allocate types of spaces (“colors”) to various urban sites, guided by personal and collective objectives. Players interact via an interface that shows a 3D map of the district, information panels, and sliders and controls to submit decisions.
[62] (Panjaburee et al., 2024)	2024	Adventure	The game uses text, images, animation, and narration to help players understand digital citizenship. It provides interactive storytelling where players make choices and see the consequences of their decisions, highlighting positive or negative behaviors.	The player must make a decision based on the story presented in the game scene.

Table 4. Cont.

Reference	Year	Game Genre	Narrative/Storyline	Game rules How the Game is Played
[70] (Tselepatiotis and Alepis, 2024)	2024	Adventure	The game shows a world full of zombies. Players are survivors who answer general knowledge questions to fight the zombies. The questions cover animals, arts, sports, history, and geography.	The player must solve the question by shooting the zombie with the right answer.
[56] (Chrysafiadi et al., 2023)	2023	Adventure	Surviving Businessman: The game has an island with castles, forests, and catacombs. The game dynamically changes the difficulty of battles and maze navigation according to the player's skills.	The player must explore mazes and fight enemies to find keys that unlock gates, allowing progression to the next stage of the game.
[74] (Felix et al., 2023)	2023	Adventure	The game narrative follows a female character named Marta through three key stages (levels of the game) of her life: childhood and adolescence, adulthood, and her search for healthcare support. 40 questions were created for the three levels of the game. This information was obtained from [77].	The player must solve quizzes.
[50] (Jiang et al., 2023)	2023	Sport	Ping pong game.	First stage of the game: The player must pick up the ping pong ball from the table and move it to the specified area to finish the task using hand movements. Competition stage: The player must try to mimic the correct striking posture and swing the arm to hit the ping pong ball.
[63] (Jondya et al., 2023)	2023	Adventure	Players undertake a journey through the game world to learn about historical events, engaging with its characters and completing missions.	The player must solve missions: visit a location, navigate the world to identify items or individuals for the successful completion of missions, search for items required by non-player characters, and answer quizzes.
[59] (Krouska et al., 2023)	2023	Logic—Puzzle	Puzzles, tasks, and missions are based on knowledge in C++ programming.	The player must solve the puzzles.
[66] (Méndez et al., 2023)	2023	Simulation	Scenarios that simulate real-world energy consumption and thermal comfort outcomes are presented to the player. Specifically, heating and cooling setpoint schedules combined with specific California locations are used as scenarios.	The player adjusts thermostat settings in the game and compares strategies to save energy. The player can use two strategies: i) using heating/cooling systems, and ii) natural ventilation, opening windows.
[69] (Rueda et al., 2023)	2023	Simulation	The game simulates real-world clinical scenarios that require electrotherapy interventions.	The player (a student) must (i) select the type and number of electrodes and position them on a virtual patient; (ii) select the current type and configure the electrical device; and (iii) select the duration and intensity of the treatment.

Table 4. Cont.

Reference	Year	Game Genre	Narrative/Storyline	Game rules How the Game is Played
[57] (Chrysafiadi et al., 2022)	2022	Adventure	FuzAd_Escape game: The player must escape from a locked house by exploring its rooms and garden, collecting objects, and using them throughout the game. Once all missions are completed, the house is unlocked. There are puzzles and quizzes on HTML programming in each room that the player must solve.	The player must solve the puzzles.
[51] (Ghorbani et al., 2022)	2022	Simulation	The game simulates daily living situations to assess cognitive functions (e.g., pattern separation and completion, visuospatial and episodic memory, decision-making ability, concentration)	Players must perform five tasks: retrieving objects' locations; memorizing objects' colors; recognizing an extra object; identifying unnatural placement of objects; and recalling the sequence of numbers.
[71] (Haryanto et al., 2022)	2022	Simulation	A disaster mitigation game that teaches what to bring in a disaster. Items include medicine, protective equipment, communication tools, and food.	The player puts items into four designated bags based on their type. Each bag must be filled. If items are placed in the wrong bag, the player loses a life. The game ends when lives run out.
[52] (Morales et al., 2022)	2022	Adventure	EtherealBreathing: Children help an “Akhi” guru to preserve the elemental balance of water, wind, earth, and fire to protect the world.	Children practice box breathing exercises to select the elements that need to be balanced in each temple of the game story
[64] (Rachmawati et al., 2022)	2022	Logic—Puzzle	A puzzle on historical figures of the Indonesian revolution.	The player must arrange the puzzle pieces properly.
[53] (Aziz Hutama et al., 2021)	2021	Action	The game shows a farmer who must catch falling apples. Each apple has a letter attached.	The player must catch the correct apple using hand movements
[54] (Lara-Alvarez et al., 2021)	2021	Action	Two game scenes: - Playing scene: The game displays 10 sum problems as falling objects that must be placed into the correct answer container before reaching the bottom. - Speaking scene: A character asks the players to speak and repeats their voice with an effect.	- Playing scene: The player must solve the sum, move the falling object (i.e., the sum), and place it into the correct container using the left, right, and down keys. - Speaking scene: The player reads a provided text, and the voice is analyzed to identify the player's emotional state.
[65] (Purnamasari et al., 2021)	2021	Logic—Puzzle	Aksara. The player must answer 10 questions shown in Latin characters by choosing the correct option. The game has a 60-s timer, and players earn a score if they finish all questions before time runs out. The countdown adds challenge to the game.	The player must answer the questions.

Table 4. Cont.

Reference	Year	Game Genre	Narrative/Storyline	Game rules How the Game is Played
[76] (Suwindra et al., 2021)	2021	Adventure	The game is based on Balinese culture: Rajapala was a hunter who married an angel, and together they had a son named I Durma. The game includes questions on emotions.	The player must answer the questions.
[68] (Björn et al., 2020)	2020	Simulation	The game presents a 3D human head model where players position EEG electrodes according to the 10–20 system.	The player must choose an electrode from a diagram showing the 10–20 electrode placement system.
[72] (Bourhim and Cherkaoui, 2020)	2020	Simulation	A virtual reality game is set in a residential building environment and includes elements such as fire, smoke, lighting, furniture, and fire-escape tools within the game scene.	The player must walk, climb, and grab objects to evacuate the building.
[58] (Chrysafiadi et al., 2020)	2020	Adventure	The game’s main character is trapped inside a building that has several rooms and a spacious garden.	The player must help the main character escape. To do so, the player must solve questions on HTML programming displayed in the game.
[61] (Lee et al., 2020)	2020	Strategy	AlphaGo game: https://deepmind.com/alphago-master-series (accessed on 28 September 2025) Players alternate placing black or white stones on a grid, aiming to gain control of more territory than their opponent.	The authors used the database of results from playing AlphaGo, the game was not played directly. The aim was to analyze the state of the game, predict outcomes (win rates), train fuzzy and machine learning models, and use these to teach students via a robot interface.
[67] (Ponce et al., 2020)	2020	Simulation	The player interacts with elements associated with a connected thermostat interface (e.g., system mode, humidity, indoor temperature, weather, quick changes, voice control, manual temperature adjustment, and menu options).	The player must solve a problem related to the thermostat’s behavior. Once solved, the stage is unlocked, and the player is rewarded.
[55] (Robles and Quintero M., 2020)	2020	Action—Race	From Arcademics: https://www.arcademics.com/ (accessed on 28 September 2025) • SpeedWay Fractions: A math-based racing game where players control a car that keeps moving forward as long as they correctly solve problems involving addition and subtraction of fractions. • SnowSprint: A math game where players move their vehicle by correctly solving multiplication problems with fractions. • Puppy Chase: A math-based racing game where players move a puppy forward if they correctly convert fractions to decimals. • Martian Hoverboards: A math-based racing game where players move forward on a hoverboard if they correctly solve expressions in the order of operations.	The player must solve the exercises

3.3. Results on RQ3—What Is the Aim of Using Fuzzy Logic in Serious Games and How Is the Fuzzy Logic System Implemented?

Based on the analysis of the studies, these can be classified into two categories regarding the architecture of the fuzzy logic system:

Fuzzy inference systems (FIS): These systems involve those that have implemented the three stages in their systems: fuzzification, fuzzy inference, and defuzzification (75%: 21 out of 28 studies).

Semi-fuzzy systems: These studies correspond to those that have only used fuzzification or fuzzy concepts (e.g., fuzzy sets). However, they did not use the three stages (25% of the studies: [57,58,60,69,72,73,75]).

3.3.1. Fuzzy Inference Systems

Fuzzy logic has been used in the studies for several purposes. Most studies in this category have applied fuzzy logic (i) to adjust game difficulty by modifying non-player character behavior (38%: 8 out of 21 studies [49,52–54,56,59,70,71]); (ii) to provide tailored feedback within the game (19%: 4 studies out of 21 [62,63,66,67]); and (iii) to compute the game score (14%: 3 out of 21 studies [64,65,74]).

Additionally, other studies have used fuzzy logic (i) to assess player features (i.e., upper limb motor function [50], player engagement [74]); (ii) to provide feedback on player performance ([55,68]); and (iii) to determine the player character [76].

In the context of fuzzification (i.e., mapping crisp values to linguistic variables), the most widely used membership functions for fuzzy input sets in the studies were triangular functions (62%: 13 out of 21 studies [49,52,53,55,59,64–67,70,71,74,76]), followed by trapezoidal functions (29%: 6 studies out of 21 [50,56,61,68,71,76]) and Gaussian functions (14%: 3 out of 21 studies [51,54,76]) (see Figure 8). Only one study [76] used and compared several membership functions for fuzzy input sets: triangular, trapezoidal, Gaussian, bell-shaped, and sigmoidal functions. Conversely, only one study [62] implemented a customized membership function for the fuzzy input sets.

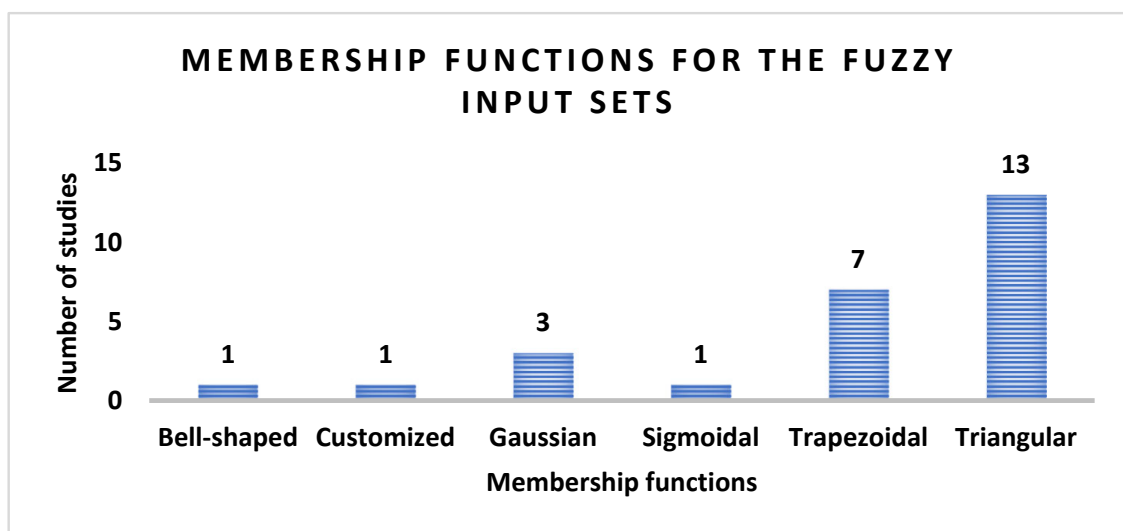


Figure 8. Membership functions for the fuzzy input sets.

It is important to note that most studies used as input to the fuzzy logic system information generated in the game, such as (i) player's skills in the game and error degree (15 out of 21 studies: [50,51,55,56,59,61–65,68,70,71,74,76]) and (ii) game score (6 out of 21 studies: [51,53,54,63,64,70]).

Conversely, few studies employed information obtained from the players as the input to the fuzzy system; for instance: (i) range of motion of hand movements detected via a Leap motion controller sensor [49], (ii) player's breathing [52], and (iii) player's emotions detected via player's voice [54].

Focusing on fuzzy inference, most studies used only one fuzzy inference system (FIS). Nevertheless, a study [50] implemented a hierarchical fuzzy inference system, which was a tree structure with three FIS subsystems. Moreover, the majority of the studies (90%: 19 out of 21 studies) employed Mamdani fuzzy inference [49–52,54–56,59,61–68,70,71,74]. In contrast, only two studies implemented Tsugeno fuzzy systems [53,76].

Figure 9 shows that the fewest number of rules used in the FIS in the studies was three rules [62], while the highest was 94 [74], followed by 81 rules [59]. The most widely used number of rules was twenty-seven [54,64,71], followed by fifteen rules [55,70] and nine rules [53,65].

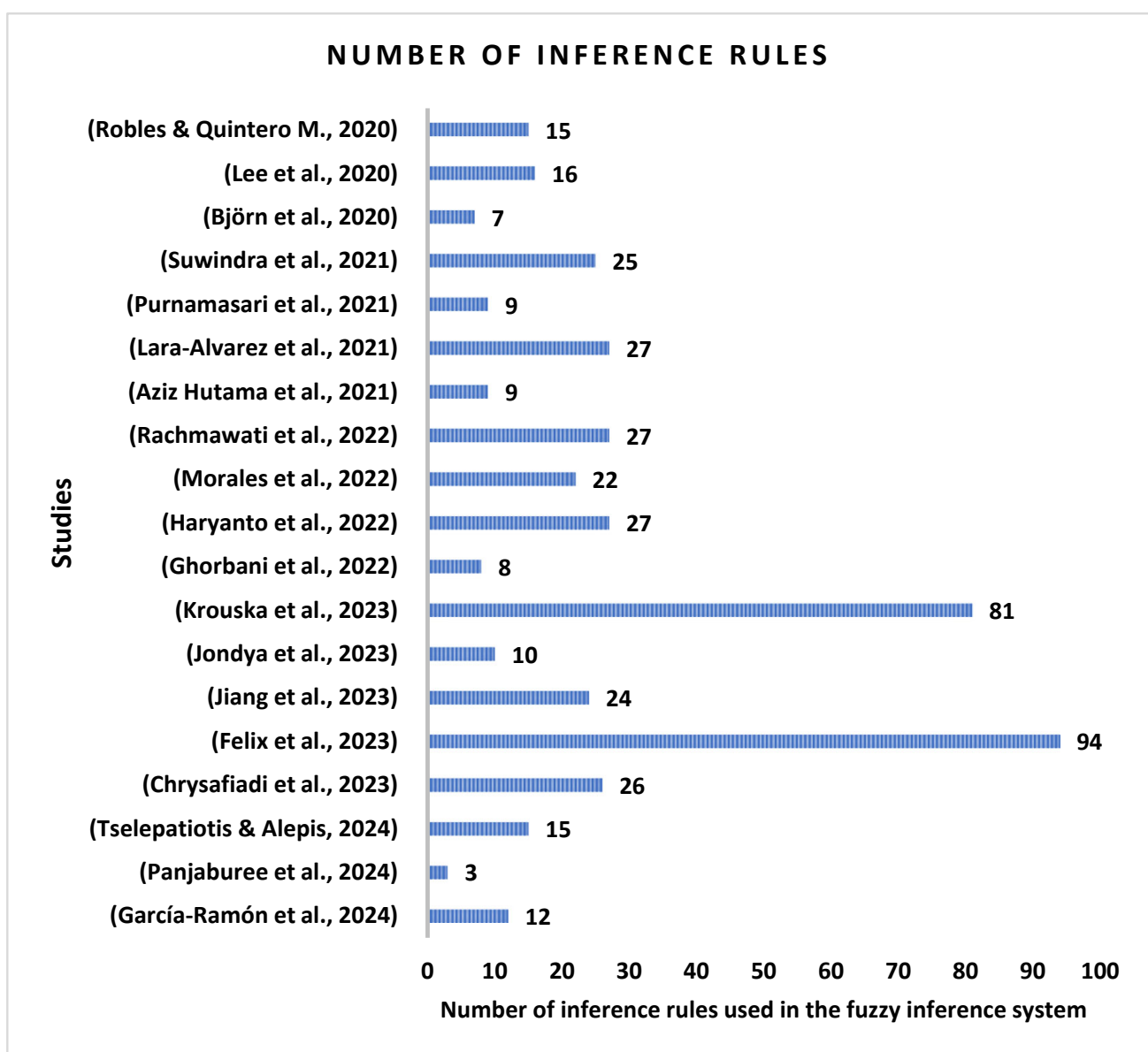


Figure 9. Number of inference rules used in the fuzzy inference system of the studies [49–56,59,61–65,68,70,71,74,76].

In the context of fuzzy logical conjunctions used in the antecedents of rules (e.g., AND, OR, NOT), that is, the “IF (conditions)” part of the rule that relates input fuzzy sets, the AND conjunction was employed in 71% of the studies (15 out of 21 studies: [50, 52–55, 59, 61, 63–66, 70, 71, 74, 76]). Only one study used AND and OR conjunctions in its rules [56]. The remaining studies did not employ conjunctions in the antecedents of the rules [49, 51, 62, 67, 68].

Focusing on the defuzzification process (i.e., converting the fuzzy output to a crisp value), most studies did not indicate the defuzzification method (10 out of 21 studies: [51–53, 56, 61, 63–65, 70, 74]). The most widely used defuzzification method in the studies was the center of area (38%: 8 out of 21 studies [50, 54, 55, 59, 66, 67, 71, 76]). Additionally, the center of sums [49, 68] and the maximum membership method [62] have been used to defuzzify. Moreover, triangular (43%: 9 out of 21 studies [50–52, 55, 59, 66, 67, 70, 74]), trapezoidal (19%: 4 out of 21 studies [49, 61, 68, 74]), singleton (2 out of 21 studies [51, 63]), and Gaussian [54] functions were used for fuzzy output sets. In most studies, fuzzy output sets were modeled using linguistic variables such as non-player character features, game appearance [49, 51–54, 56, 66, 67, 70, 71], and player performance [59, 62, 63] to adjust game difficulty. Other studies employed the fuzzy output sets to assess the player’s performance in terms of linguistic variables [50, 55, 61, 65, 68, 74].

Further details on the fuzzy logic systems in these studies can be found in Table 5.

3.3.2. Semi-Fuzzy Systems

A quarter of the studies included in this review implemented semi-fuzzy systems [57, 58, 60, 69, 72, 73, 75]. Most studies in this category used fuzzy sets. Specifically, a study [60] employed fuzzy sets, fuzzy relations, and fuzzy attribute implications to represent the students’ solution in the light-bot game. Other studies used fuzzy sets: (i) to model the knowledge level of students [57, 58]; (ii) to model perception of a non-player character in terms of distance to the player in the game scene [73]; (iii) to assess criteria for virtual environment usability [72]; and (iv) to represent values of electrode placement (well oriented, centered, and well distributed) and electrical parameters for the treatment (i.e., pulse width, frequency, amplitude, rest time, treatment time) [69].

Finally, only one study [75] employed fuzzy AND aggregation to compute scores for urban layouts (spatial configurations) in a 3D urban design space. Further details of these studies can be found in Table 6.

3.4. Results on RQ4—Were Experiments Conducted in the Studies?

Most studies conducted experiments involving people participation (79%: 22 out of 28 studies). The smallest sample size in the studies was five participants [53], while the largest sample size was 206 participants [55] (see Figure 10). Most studies (77%: 17 out of 22 studies) used samples of more than 30 participants, while only a few studies included fewer than 30 participants [52, 53, 65, 70, 76]. Conversely, a few studies did not indicate the number of volunteers who participated in the experiments [60, 67, 75]. Additionally, a few studies conducted experiments using simulations that did not involve the participation of people [66, 71]. Some studies used the game as an assistive tool. Of these, only two conducted experiments with participants who had the medical condition that the game was designed to address (i.e., hemiplegia [50], autism [52]).

Table 5. Fuzzy inference systems used in serious games.

Reference	Year	Aim	Fuzzification	Fuzzy Inference	Defuzzification
[49] (García-Ramón et al., 2024)	2024	To adjust the game difficulty (i.e., the position and size of the box to be destroyed by the player) according to the players' range of motion.	Fuzzy input sets Range of motion (ROM) of the hand movements: • Ulnar movement: {low, medium, high} • Extension movement: {low, medium, high} • Radial movement: {low, medium, high} • Flexion movement: {low, medium, high} Membership functions: All triangular	Mamdani fuzzy inference system 12 rules in total: Adjustment of the box position: • Three rules with the structure: IF (ulnar ROM) THEN box position • Three rules with the structure: IF (extension ROM) THEN box position Adjustment of the box size: • Three rules with the structure: IF (radial ROM) THEN box size • Three rules with the structure: IF (flexion ROM) THEN box size	Fuzzy output sets Game difficulty: • Box position: {easy, medium, hard} • Box size: {easy, medium, hard} Membership functions: All trapezoidal Defuzzification method: center of sums
		The fuzzy logic system was implemented using C# in Unity version 2021.3.24f1.			
[62] (Panjaburee et al., 2024)	2024	To offer tailored feedback on digital citizenship behaviors in the game.	Fuzzy input sets Error degree is computed using max-min composition on: (i) response of a sheet table; (ii) the test items covered in the storytelling scene; and (iii) record of the storingtelling scenes related to each aspect of digital citizenship • error_degree: {low, average, high} Membership functions: For Low: • IF the error degree (X) is 0, THEN low = 1 • IF $0 < X < 0.5$, THEN low = $1 - 2X^2$ • IF $0.5 \leq X < 1$, THEN low = $2(X - 1)^2$ • IF $X = 1$, THEN low = 0 For Average: • IF $0 \leq X < 0.25$ THEN average = $2(X/0.5)^2$ • IF $0.25 \leq X < 0.75$, THEN average = $1 - 2(X - 0.5/0.5)^2$ • IF $X = 0$ or $X = 1$, THEN average = 0.	Mamdani fuzzy inference system three rules in total with the following structure: • IF error_degree THEN digital citizenship behavior_status	Fuzzy output sets • Digital citizenship behavior_status: {poorly performed, partially performed, well performed} Defuzzification method: The maximum membership between poorly performed, partially-performed, and well-performed

Table 5. Cont.

Reference	Year	Aim	Fuzzification	Fuzzy Inference	Defuzzification
			For High: - IF $0 \leq X < 0.5$, THEN high = $2(X^2)$ - IF $0.5 \leq X < 1$, THEN high = $1 - (2X - 1)^2$ - IF $X = 0$, THEN high = 0 - IF $X = 1$, THEN high = 1 Information on the fuzzy sets was obtained from [78].		
[70] (Tselepatiotis and Alepis, 2024)	2024	To adjust zombie behavior according to player skill: more aggressive for skilled players and easier for less skilled players.	Fuzzy input sets - Player skill level: {Bad, Normal, Good}—Triangular membership functions - Player overall score: {Low, Normal, High}—Triangular membership functions - Number of zombies: {Zero, Low, Normal, High}—The membership function is not indicated	Mamdani fuzzy inference system 15 rules in total: Initial three fuzzy rules with the following structure: - IF player skill level THEN zombies Three rules with the structure: - IF player overall score THEN Player. Three rules with the following structure: - IF player skill level THEN zombies' speed One rule with the following structure: - IF number of zombies is Zero THEN the zombies' level will be [inert, inert, inert] Five rules with the following structure: - IF number of zombies AND player score THEN zombies' level	Fuzzy output sets - Zombies {Fewer, Normal, More} The membership function is not indicated. - Player {Bad, Normal, Good}—The membership function is not indicated. - Zombie's speed {Slow, Normal, Fast}—Triangular membership functions - Zombie level {Aggressive, alert, inert}—The membership function is not indicated. There are states: - Aggressive zombies chase the player using the A* algorithm. - Alert zombies chase the players when they are in their field of view. - Inert zombies do not chase the players even when they are in their field of view. The defuzzification method is not indicated

Table 5. Cont.

Reference	Year	Aim	Fuzzification	Fuzzy Inference	Defuzzification
[56] (Chrysafiadi et al., 2023)	2023	To adapt dynamically the game difficulty (i.e., the number and difficulty of the questions and quizzes).	Fuzzy input sets	Mamdani fuzzy inference system 26 rules in total: Battle skill (10 rules). Examples: • IF loss of life = extremely large THEN battle stereotype = beginner • IF length of battle = very fast AND loss of life = slow THEN battle stereotype = beginner • IF loss of life = significant AND length of battle = very fast OR length of battle = fast OR length of battle = slow THEN battle stereotype = easy	Fuzzy output sets • Battle Stereotype: {beginner, easy, normal, hard, expert}—numeric ranges for time to beat the enemy and for the damage caused by enemies to the player. • Maze Stereotype: {good, moderate, novice}—numeric ranges for time to navigate in the maze, damage caused by traps to the player, and number of available help requests.
			For Battle skill: • Length of time each battle has taken: {very fast, fast, slow, very slow} • Loss of life by the enemy: {very little, little, medium, significant, extremely large} For skill in maze navigation: • Time spent navigating the maze: {very fast, fast, slow, very slow} • Loss of life suffered from maze traps: {little, medium, large} • Number of times the player requested help to continue navigating: {minor, moderate, much, extended} Membership functions: All trapezoidal	Maze navigation skill (16 rules). Examples: • IF loss of life = large AND request help = extended THEN maze stereotype = novice • IF time spent = very slow OR time spent = slow AND loss of life = large THEN maze stereotype = novice • IF time spent = very slow AND loss of life = medium AND request help = moderate OR request help = much OR request help = extended THEN maze stereotype = novice	The defuzzification method is not indicated

Table 5. Cont.

Reference	Year	Aim	Fuzzification	Fuzzy Inference	Defuzzification
[74] (Felix et al., 2023)	2023	Two fuzzy systems: - Fuzzy System Exploration Type (FSE): To check whether the player is engaging with the components of the challenge. - Fuzzy Performance Assessment System (FPAS): To compute the player's final performance.	Fuzzy input sets for FSE: - TimeChallenge: {insufficient, little adequate, adequate, long} - TimeQuestion: {insufficient, little adequate, adequate, long} - TimeFeedback: {insufficient, adequate, not suitable, far_away} All triangular membership functions	Mamdani fuzzy inference system FSE: 64 rules: The structure of the rules is not indicated	Fuzzy output sets for FSE: Player's exploration: {very fast, fast, poorly adequate, slow, very slow}—Triangular membership functions.
			Fuzzy input sets for FPAS: - Player's performance: {very bad, bad, regular, good, very good} - Player's exploration (this is computed using FSE): {very fast, fast, poorly adequate, slow, very slow} All triangular membership functions	FPAS: 30 rules with the following structure: IF player's performance AND player's exploration THEN player's knowledge	Fuzzy output sets for FPAS: Player's knowledge: {insufficient, reasonable, good}—Trapezoidal membership functions. The defuzzification method is not indicated
[50] (Jiang et al., 2023)	2023	To evaluate upper limb motor function in stroke patients.	Fuzzy input sets - Movement variance: difference between the player's reported hand movement path and the target path. - Movement length ratio: ratio between the reported hand movement distance during the preparation phase and the straight-line distance from the starting point to the endpoint. - Remaining time: the remaining time following the player's completion of the target task during the preparation stage. - Maximum stretch ratio: The system requests the player to move the ball within a defined area, involving arm elevation, during which the system records the relative height of the player's elbow to shoulder and hand to elbow.	Hierarchical fuzzy inference A tree structure with three FIS subsystems Mamdani fuzzy inference system 24 rules in total: FIS 1: To identify if the player corresponds to the healthy group or the hemiparesis group Eight rules with the structure: IF remaining time AND average hand speed AND average elbow angular speed THEN hemiparesis, control	Fuzzy output sets FIS 1: - Hemiparesis: {possible, impossible} - Control: {possible, impossible} FIS 2: - Severe: {possible, impossible} {Moderate, mild}: {possible, impossible}

Table 5. Cont.

Reference	Year	Aim	Fuzzification	Fuzzy Inference	Defuzzification
			FIS (Fuzzy Inference System) 1: - Remaining time: {low, high} - Average hand speed: {low, high} - Average elbow angular speed: {low, high} FIS 2: - Movement variance: {high, low} - Movement length ratio: {low, high} - Remaining time: {low, high} FIS 3: - Movement length ratio: {low, high} - Maximum stretch ratio: {low, high} - Average hand speed: {low, high} Membership functions: All trapezoidal	FIS 2: To identify if the player corresponds to the {moderate, mild} hemiparesis group or the severe hemiparesis group Eight rules with the following structure: - IF movement variance AND movement length ratio AND remaining time THEN severe, {moderate, mild} FIS 3: to identify if the player corresponds to the mild hemiparesis group or the moderate hemiparesis group Eight rules with the following structure: - IF movement length ratio AND maximum stretch ratio AND average hand speed THEN moderate, mild	FIS 3: - Moderate: {possible, impossible} - Mild: {possible, impossible} Membership functions: All triangular Defuzzification method: Center of area
[63] (Jondya et al., 2023)	2023	To adapt gameplay based on the player's understanding of the historical narrative and evaluate that understanding dynamically.	Fuzzy input sets - Player's average score: {poor, decent, great}—Trapezoidal membership functions. - Time spent in finishing the assessment: {slow, fast}—The membership function is not indicated.	Mamdani fuzzy inference system 10 rules: IF player's average score AND time spent THEN player's understanding	Fuzzy output sets Player's understanding {0: default—first assessment, 1: poor understanding, 2: decent understanding, 3: great understanding}—Singleton membership functions. The defuzzification method is not indicated
[59] (Krouska et al., 2023)	2023	To estimate the level of challenge in the game.	Fuzzy input sets - Student progress: {Low, Intermediate, Advanced} - Gaming time: {start, middle, end} - Student knowledge: {novice, intermediate, advanced} All triangular membership functions	Mamdani fuzzy inference system 81 rules: IF student progress AND gaming time AND student knowledge THEN challenging level	Fuzzy output sets Challenging level: {easy, intermediate, difficult}—Triangular membership functions. Defuzzification method: center of area

Table 5. Cont.

Reference	Year	Aim	Fuzzification	Fuzzy Inference	Defuzzification
[66] (Méndez et al., 2023)	2023	To classify players based on their personality traits (openness, conscientiousness, extraversion, agreeableness, and neuroticism).	Fuzzy input sets	Mamdani fuzzy inference system	Fuzzy output sets
		To use the “effects” obtained from the first system (engagement, energy usage, attitude, and knowledge) to propose the most suitable gamification elements (e.g., challenges, points, badges, feedback) for tailoring the interface.	First part of the fuzzy logic decision system Based on personality traits: • Openness: {low, medium, high} • Conscientiousness: {low, medium, high} • Extraversion: {low, medium, high} • Agreeableness: {low, medium, high} • Neuroticism: {low, medium, high} The membership function is not indicated	First part of the fuzzy logic decision system The number of rules is not indicated. Only two rules are provided as examples with the following structure: • IF openness AND conscientiousness AND extraversion AND agreeableness AND neuroticism THEN engagement, energy usage, attitude, knowledge, home-focused, traditionalist cost-focused, philanthropist, socializer, free spirit, achiever, player, disruptor • IF openness AND conscientiousness AND extraversion AND agreeableness AND neuroticism THEN green advocate, engagement, energy usage, attitude, knowledge, philanthropist, socializer, free spirit, achiever, player, disruptor Second part of the fuzzy logic decision system The number of rules is not indicated. • IF effect elements {Engagement. Energy usage, Attitude, Knowledge} THEN trigger elements, interface elements, rewards elements	First part of the fuzzy logic decision system Based on effect elements: • Engagement: {low, medium, high} • Energy usage: {low, medium, high} • Attitude: {low, medium, high} • Knowledge: {low, medium, high} Membership functions: All triangular Based on energy end-user elements: • Green advocate: {active} • Traditionalist cost-focused: {active} • Home focused: {active} • Non-green selective: {} • Disengaged: {} The membership function is not indicated, and the linguistic variables are not fully indicated. The linguistic variables identified are based on the two rules presented in the article. Based on gamified player elements: • Philanthropist: {high} • Socializer: {high} • Free spirit: {high} • Achiever: {low, high} • Player: {low, high} • Disruptor: {low} The membership function is not indicated, and the linguistic variables are not fully indicated. The linguistic variables identified are based on the two rules presented in the article.

Table 5. Cont.

Reference	Year	Aim	Fuzzification	Fuzzy Inference	Defuzzification
					Second part of the fuzzy logic decision system Based on trigger elements: - Challenges: {low, medium, high} - Social comparison: {low, medium, high} - Competition: {low, medium, high} Based on interface elements: - Dashboard: {low, medium, high} - Monitoring: {low, medium, high} - Feedback: {low, medium, high} - Progress bar: {low, medium, high} - Leaderboard: {low, medium, high} Based on rewards elements: - Points: {low, medium, high} - Badges: {low, medium, high} - Prizes: {low, medium, high} - Coupons: {low, medium, high} - Bill discounts: {low, medium, high} Membership functions: All triangular Defuzzification method: center of area
[51] (Ghorbani et al., 2022)	2022	To assist players in making decisions and to send reminders of events through messages based on their cognitive state.	Fuzzy input sets Variables on players' real-time location, and their cognitive state: - Game score: {low, high}—Gaussian membership functions. - Movement: {low, medium, high}— Gaussian membership functions. - Time: {early morning, morning, early afternoon, late afternoon, evening, night, midnight}—The membership function is not indicated. - Gas detection: {yes, no}—The membership function is not indicated. - Distance: {near, far, very far}—The membership function is not indicated.	Mamdani fuzzy inference system Only eight rules were provided: - IF time is evening THEN game is started - IF time is morning THEN voice message is 1 AND image message is 1 - IF gas detection is yes THEN relay status is yes AND voice message is 2 - IF distance is near THEN voice message is 3 AND image message is 2 - IF distance is near THEN voice message is 4 AND image message is 3	Fuzzy output sets Messages to the players: - Game: {start, stop}—The membership function is not indicated. - Voice message: {1, 2, 3, 4, 5, 6}—Singleton membership functions - Reminder: {no, yes}—The membership function is not indicated. - Relay status: {yes, no}—Triangular membership functions. - Image message: {1, 2,...}—Singleton membership functions.

Table 5. Cont.

Reference	Year	Aim	Fuzzification	Fuzzy Inference	Defuzzification
			- Humidity: {very dry, dry, humid, very humid}—The membership function is not indicated. - Temperature: {very cold, cold, cool, mild, warm, hot, very hot}—The membership function is not indicated	- IF time is early afternoon THEN voice message is 4 AND image message is 4 - IF time is early afternoon THEN voice message is 6 - IF game score is high THEN reminder is no	The defuzzification method is not indicated
[71] (Haryanto et al., 2022)	2022	To create the dynamic behavior of how often the item appears in the game.	Fuzzy input sets - Time constraint (TIME): {short, moderate, long}—Trapezoidal membership functions. - Health point (HP) as limited resources component: {low, medium, high}—Trapezoidal membership functions. - FAIL, the number of failed attempts: {rare, occasionally, often}—Triangular membership functions.	Mamdani fuzzy inference system 27 rules in total: IF health point AND time constraint AND fail THEN (ITEM TYPE): AddHP, addTime, Slow (add delay to the timer), double (double the score obtained)	Fuzzy output sets Item occurrence frequency: - Add_time: {few, plenty} - Add_healthPoint: {few, plenty} - DelayToTheTimer: {few, plenty} - DoubleTheScore: {few, plenty} Membership functions: All shoulders Defuzzification method: center of area
[52] (Morales et al., 2022)	2022	To adjust the game difficulty according to the breathing prompts and visual attention.	Fuzzy input sets - Percentage of auditory prompts—indications—needed: {few, average, many} - Percentage of breaths: {very good, good, normal, bad, very bad} Membership functions: All triangular	Mamdani fuzzy inference system 22 rules: Defined by experts: a pediatrician and an author IF auditory prompts AND breaths THEN game difficulty	Fuzzy output sets Game difficulty: {very easy, easy, normal, hard, very hard} Membership functions: triangular membership functions The defuzzification method is not indicated
[64] (Rachmawati et al., 2022)	2022	To compute game scores and to measure game time.	Fuzzy input sets - Game level score: {small, medium, many} - Game time: {less, middle, good} - Number of correct answers: {a little, enough, good} Membership functions: All triangular	Mamdani fuzzy inference system 27 rules: IF game level AND game time AND number of correct answers THEN (the consequence is not indicated)	Not explained

Table 5. Cont.

Reference	Year	Aim	Fuzzification	Fuzzy Inference	Defuzzification
[53] (Aziz Hutama et al., 2021)	2021	To adjust the game difficulty dynamically. The fuzzy logic system was implemented using C#.	Fuzzy input sets - Life: {few, medium, many} - Game score: {low, medium, high} Membership functions: All triangular	Zero-order Sugeno fuzzy model Nine rules with the following structure: IF game score AND life THEN difficulty	Fuzzy output sets - Difficulty: {level_down, stay_on_level, level_up}—constant variables: - Level_down = 1 - Stay_on_level = 2 - Level_up = 3 The defuzzification method is not indicated
[54] (Lara-Alvarez et al., 2021)	2021	To assess players' performance and emotions to adjust the difficulty and visual design of the game. The player's emotional state (which is modeled using Russell's circumplex model) is detected via voice.	Fuzzy input sets - Score ratio (success rate): {low, medium, high} - Current player's emotional state (arousal: {low, medium low, medium, high, very high}, valence {pleasant, unpleasant}) - Nominal emotion induced by the aesthetics of the previous stage: (arousal {low, medium low, medium, high, very high}, valence {pleasant, unpleasant}) Membership functions: All Gaussian	Mamdani fuzzy inference system 27 rules in total: Three rules with the following structure: - IF score ratio THEN difficulty level Twenty-four rules with the following structure: - IF current player emotional state AND nominal emotion THEN expected emotion induced (arousal, valence)	Fuzzy output sets - Difficulty level change factor: {decrease, equal, increase} - Arousal emotion: {low, medium low, medium, high, very high} - Valence emotion: {pleasant, unpleasant} Membership functions: All Gaussian Defuzzification method: center of area
[65] (Purnamasari et al., 2021)	2021	To compute the game score.	Fuzzy input sets - Time (time taken to respond to questions): {small duration, medium duration, many duration} - Correct answer: {less, medium, good} Membership functions: All triangular	Mamdani fuzzy inference system Nine rules with the following structure: IF time AND number of correct answers THEN score	Fuzzy output sets Score: {bad, enough, excellent} The membership function and the defuzzification method are not indicated.
[76] (Suwindra et al., 2021)	2021	To determine the player character. The fuzzy logic system was implemented using Simulink of the Fuzzy Logic Toolbox for MATLAB 2021.	Fuzzy input sets - Game-factor (F): player's response to the game. - Emotional input (E): player's emotion. The membership functions were adjusted using backpropagation. Eight types were analyzed: triangular, trapezoidal, generalized bell-shaped, gaussian, two Gaussian, sigmoidal, difference sigmoidal, and product sigmoidal.	Sugeno neuro-fuzzy system 25 rules in total: IF F AND E THEN Z	Fuzzy output sets Z is a player character with the character aspects independence, resilience, discipline, religion, honest, tolerant, patriotic, creative, integrity, and friendly. Defuzzification method: center of area

Table 5. Cont.

Reference	Year	Aim	Fuzzification	Fuzzy Inference	Defuzzification
[68] (Björn et al., 2020)	2020	To provide human feedback (i.e., a directional arrow indicating where the electrode should be moved and a linguistic variable—word describing the required magnitude of movement) on the correct position of the electroencephalography electrode placement. The fuzzy logic system was implemented using C#.	Fuzzy input sets Distance (error between the correct and the given positions of the electroencephalography electrode): {perfect, very close, close, medium, far medium, far, far away}—Trapezoidal membership functions. Information on the fuzzy system was obtained from [79].	Mamdani fuzzy inference system Seven rules with the following structure: IF distance THEN feedback	Fuzzy output sets - Human feedback on user interface consists of (i) a directional arrow indicating where the electrode should be moved, and (ii) a linguistic variable describing the required magnitude of movement. - Feedback: {exact, a little bit more, a bit more, somewhat more, more, much more, very much more }—Trapezoidal membership functions. Defuzzification method: center of sums
[61] (Lee et al., 2020)	2020	To predict the win rate of the game of Go as Black or White using a genetic algorithm as well. Fuzzy Markup Language was used in the fuzzy logic system.	Fuzzy input sets For the number of simulations of Black and White: - DSN: {low, high} For the win rates of Black and White: - DTMR: {low, high} For the best-move matching rates of Black and White: - DWR: {low, med_low, med_high, high} The membership functions are not indicated; however, it can be inferred that they are trapezoidal functions.	Mamdani fuzzy inference system 16 rules: IF DSN AND DTMR AND DWR THEN EWR	Fuzzy output sets For the win rates of Black and White: EWR: {low, med, high} The membership functions are not indicated; however, it can be inferred that they are trapezoidal functions. The defuzzification method is not indicated

Table 5. Cont.

Reference	Year	Aim	Fuzzification	Fuzzy Inference	Defuzzification
[67] (Ponce et al., 2020)	2020	To identify the gamification and game elements to be shown in the interface that best match each energy consumer type. The fuzzy logic system was implemented using LabVIEW 2018.	Fuzzy input sets Effects elements: - Engagement: {low, medium, high} - Energy usage: {low, medium, high} - Attitude: {low, medium, high} - Knowledge: {low, medium, high} Membership functions: All triangular	Mamdani fuzzy inference system The number of rules is not indicated - IF energy usage THEN challenges - IF energy usage THEN competition - IF energy usage THEN dashboard - IF energy usage THEN monitoring - IF energy usage THEN coupons - IF energy usage THEN bill discounts - IF engagement THEN challenges - IF engagement THEN competition - IF engagement THEN progress bar - IF engagement THEN leaderboard - IF engagement THEN points - IF engagement THEN badges - IF engagement THEN prizes - IF attitude THEN social comparison - IF attitude THEN leaderboard - IF attitude THEN badges - IF knowledge THEN challenges - IF knowledge THEN dashboard - IF knowledge THEN monitoring - IF knowledge THEN feedback - IF knowledge THEN prizes	Fuzzy output sets Trigger elements: - Challenges: {low, medium, high} - Social comparison: {low, medium, high} - Competition: {low, medium, high} Interface elements: - Dashboard: {low, medium, high} - Monitoring: {low, medium, high} - Feedback: {low, medium, high} - Progress bar: {low, medium, high} - Leaderboard: {low, medium, high} Rewards elements: - Points: {low, medium, high} - Badges: {low, medium, high} - Prizes: {low, medium, high} - Coupons: {low, medium, high} - Bill discounts: {low, medium, high} Membership functions: All triangular Defuzzification method: center of area
				Mamdani fuzzy inference system 15 rules in total: Eight rules with the following structure: - IF playtime AND correct answers THEN player performance Seven rules with the following structure: - IF playtime AND incorrect answers THEN player performance	Fuzzy output sets Player performance: {beginner, medium, pro}—Triangular membership functions. Defuzzification method: center of area

Table 6. Semi-fuzzy systems used on serious games.

Reference	Year	Aim	Fuzzy Foundation
[60] (Guniš et al., 2025)	2025	To analyze and interpret students’ solutions to the Light-Bot game, not directly within the game mechanics itself. Fuzzy attribute implications were implemented using the R package fcaR 1.2.1.9000.	Fuzzy sets, fuzzy relations, and fuzzy attribute implications were used to express the students’ solutions. Attributes of the students’ solutions (e.g., command count, strategy similarity) were normalized in [0, 1] to reflect degrees of attribute satisfaction.
[73] (Liu et al., 2024)	2024	To model the perception of a virtual character in terms of distance from a stimulus.	The perception $P(t)$ was modelled using fuzzy sets as follows: $P(t) = \begin{cases} 0 & : d_i > d_{bi} \\ fl \left(\frac{d_{ai} - d_i}{d_{ai} - d_{bi}} \right) & : d_{ai} \leq d_i \leq d_{bi} \\ fl & : d_i < d_{ai} \end{cases}$ where fl is the intensity of the stimulus, $fl \in [0, 1]$ d_i distance between the virtual character and the stimulus d_{ai} shortest distance d_{bi} longest distance
[75] (Nourian et al., 2024)	2024	To assign a multi-criteria score to each voxel based on urban layouts (spatial configurations) in a 3D urban design space by aggregating multiple performance indicators (e.g., daylight access, walkability, accessibility, and environmental quality) in a way that reflects real-world ambiguity, trade-offs, and stakeholder preferences. The datasets of the study are available in the EquiCityData repository: https://github.com/shervinazadi/EquiCity_Data, (accessed on 20 August 2025).	It uses a fuzzy AND aggregation to compute a score for each voxel (volumetric cell) based on multiple quality criteria.

Table 6. Cont.

Reference	Year	Aim	Fuzzy Foundation
[69] (Rueda et al., 2023)	2023	To represent the values of electrode placement (well oriented, centered, and well distributed) and electrical parameters for the treatment (i.e., pulse width, frequency, amplitude, rest time, treatment time) using fuzzy sets.	Fuzzy sets for:
			- Electrode placement (orientation, centering, and distribution): - Correctly oriented using: Angle between electrodes: {crosswise, lengthwise}—left/right shoulders membership functions. - Correctly centered using: Distance between the midpoint of the electrodes and the center of the ellipsoid of an anatomical zone (body area): {centered}—left shoulder membership functions. - Correctly distributed using: Spacing between the electrodes: {well distributed}—trapezoidal membership functions. - Electrical parameters on current: - Pulse width (milliseconds): {Tens(transcutaneous electrical nerve stimulation) asymmetric biphasic, TENS (transcutaneous electrical nerve stimulation) symmetric biphasic}—trapezoidal membership functions; and {Trabert}—discrete - Rest time (milliseconds): {Trabert}—triangular membership functions. - Frequency (Hertz): {Trabert, Tens(transcutaneous electrical nerve stimulation) asymmetric biphasic, TENS (transcutaneous electrical nerve stimulation) symmetric biphasic}—trapezoidal membership functions, and {Interferential bipolar; Interferential tetrapolar; Interferential (Russian stimulation)}—discrete. - Amplitude modulation frequency (Hertz): {Interferential bipolar; Interferential tetrapolar}—trapezoidal membership functions; and {Interferential (Russian stimulation)}—triangular membership functions - Treatment time (minutes): {Galvanic direct/interrupted; Iontophoresis; Dyadinamic DF/MF/LP/CP/RS; Trabert; TENS with asymmetric biphasic waveform; Interferential bipolar; Interferential tetrapolar; Interferential (Russian stimulation)}—trapezoidal membership functions; and {TENS with symmetric biphasic waveform}—triangular membership functions.

Table 6. Cont.

Reference	Year	Aim	Fuzzy Foundation
[57] (Chrysafiadi et al., 2022)	2022	To create a profile of the player based on their quiz performance and determine whether it is necessary to extend the learning experience by adjusting the storyline and introducing additional exercises.	Fuzzy sets are used to represent the knowledge level of each student: {Beginner, Moderate, Good, Expert}, and these are mapped to cognitive states {Does not learn, Learns, Forgets, Reaches the target knowledge} using rules and state machines. $\mu_{\text{Beginner}} = \begin{cases} 1, & x < 40 \\ 1 - \frac{x-40}{5}, & 40 < x < 45 \\ 0, & x \geq 45 \end{cases}$ $\mu_{\text{Moderate}} = \begin{cases} \frac{x-40}{5}, & 40 < x < 45 \\ 1, & 45 \leq x \leq 55 \\ 1 - \frac{x-55}{10}, & 55 < x < 65 \\ 0, & x \leq 40 \vee x \geq 65 \end{cases}$ $\mu_{\text{Good}} = \begin{cases} \frac{x-55}{10}, & 55 < x < 65 \\ 1, & 65 \leq x \leq 80 \\ 1 - \frac{x-80}{5}, & 80 < x < 85 \\ 0, & x \leq 55 \vee x \geq 85 \end{cases}$ $\mu_{\text{Expert}} = \begin{cases} \frac{x-80}{5}, & 80 < x < 85 \\ 1, & 85 \leq x \leq 100 \\ 0, & x \leq 80 \end{cases} \text{ where } x \text{ is the learner's degree in the quizzes.}$
[72] (Bourhim and Cherkaoui, 2020)	2020	To assess the virtual environment usability.	Layers were used to represent the criteria (e.g., interaction, side effects, engagement, navigation, object manipulation, visual criteria, auditory criteria, presence, immersion) assessed for usability. Each layer represents a criterion. Four fuzzy sets were used to assess each criterion: {high, relatively high, moderate, low}.
[58] (Chrysafiadi et al., 2020)	2020	To adjust the game scene dynamically according to each learner's individual needs.	Fuzzy sets are used to represent the student knowledge level: {novice, average, good, excellent} on HTML programming. They use trapezoidal membership functions. These fuzzy sets serve as fuzzy states within a state diagram that shows how the game adapts dynamically and tracks the learner's progression over time.

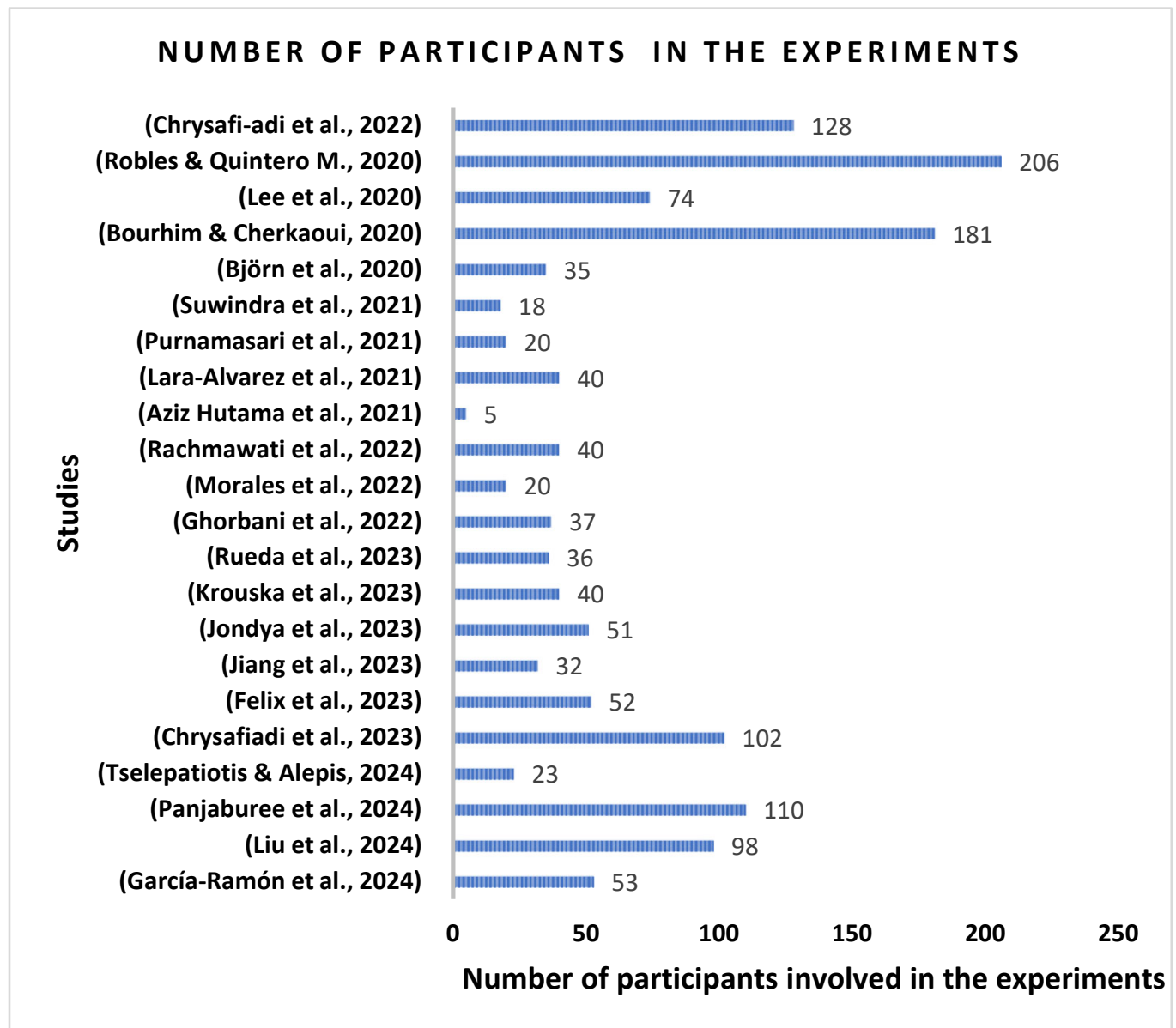


Figure 10. Number of participants involved in the experiments of the studies [49–57,59,61–65,68–70,72–74,76].

Instruments and metrics were used in most studies to assess their proposals (93%: 26 out of 28 studies). Most of them used questionnaires (58%: 15 out of 26 studies), followed by game score (38%: 10 out of 26 studies) as the instruments and metrics—see Figure 11. Other metrics included players' aspects such as player's solution [60], player's response time [54], player's emotion computed via electroencephalography signal [49], player's visual attention computed via an eye tracker [62], player's breathing [52], player's assessment from an expert [69], test on the HTML programming language [57], MoCA test for identifying mild cognitive impairments [51], and Fugl–Meyer assessment to measure motor function in the upper extremity [50]. Other studies employed metrics related to the system implemented in the game: output of the fuzzy system [64,66,71] and accuracy from classifiers [50].

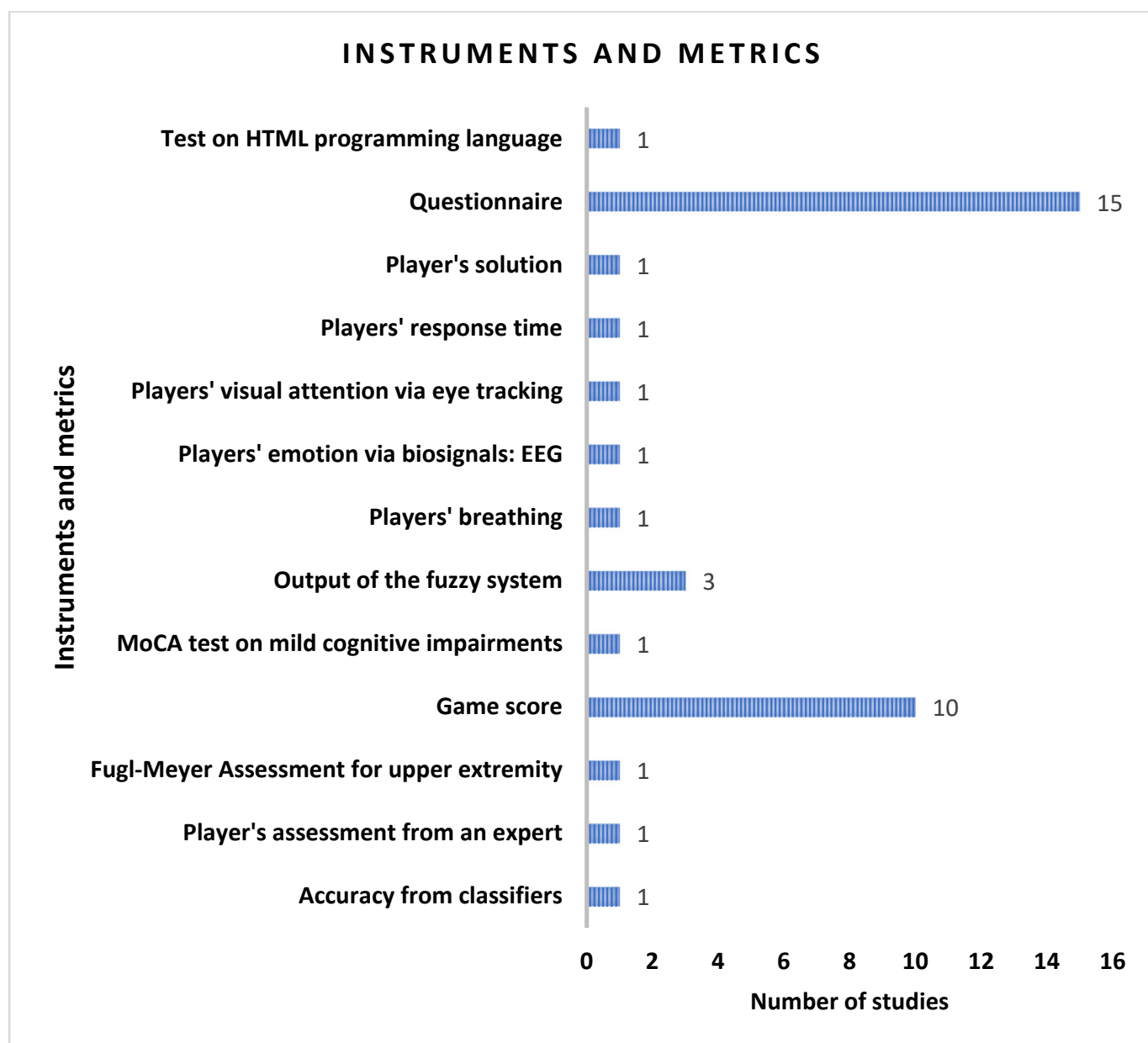


Figure 11. Instruments and metrics used in the studies.

A key aspect of assessing the reliability of the findings in the studies is to conduct statistical analysis, as it helps ensure that results are not due to random variation. In this regard, it can be seen from Figure 12 that over half of the studies did not conduct a statistical analysis (57%: 16 out of 28 studies). Conversely, the most widely used statistical tests were the (i) *t*-test, which is a parametric statistical method used to assess whether there is a significant difference in a study variable between two treatment groups [51,54,56,57]; (ii) Wilcoxon signed-rank test, which is a non-parametric statistical test that uses ranks to determine whether there is a significant difference in a study variable between two paired treatment groups [49,54,68]; and (iii) Pearson correlation coefficient, which measures how strongly two study variables are related in a linear relationship [51,72,76]. Other statistical tests used to analyze the results were Spearman correlation, Shrout and Fleiss intraclass correlation, Cronbach's alpha, Cohen's Kappa coefficient, Shapiro–Wilk, Kruskal–Wallis, Fisher's exact test, Chi-square test, and ANCOVA test.

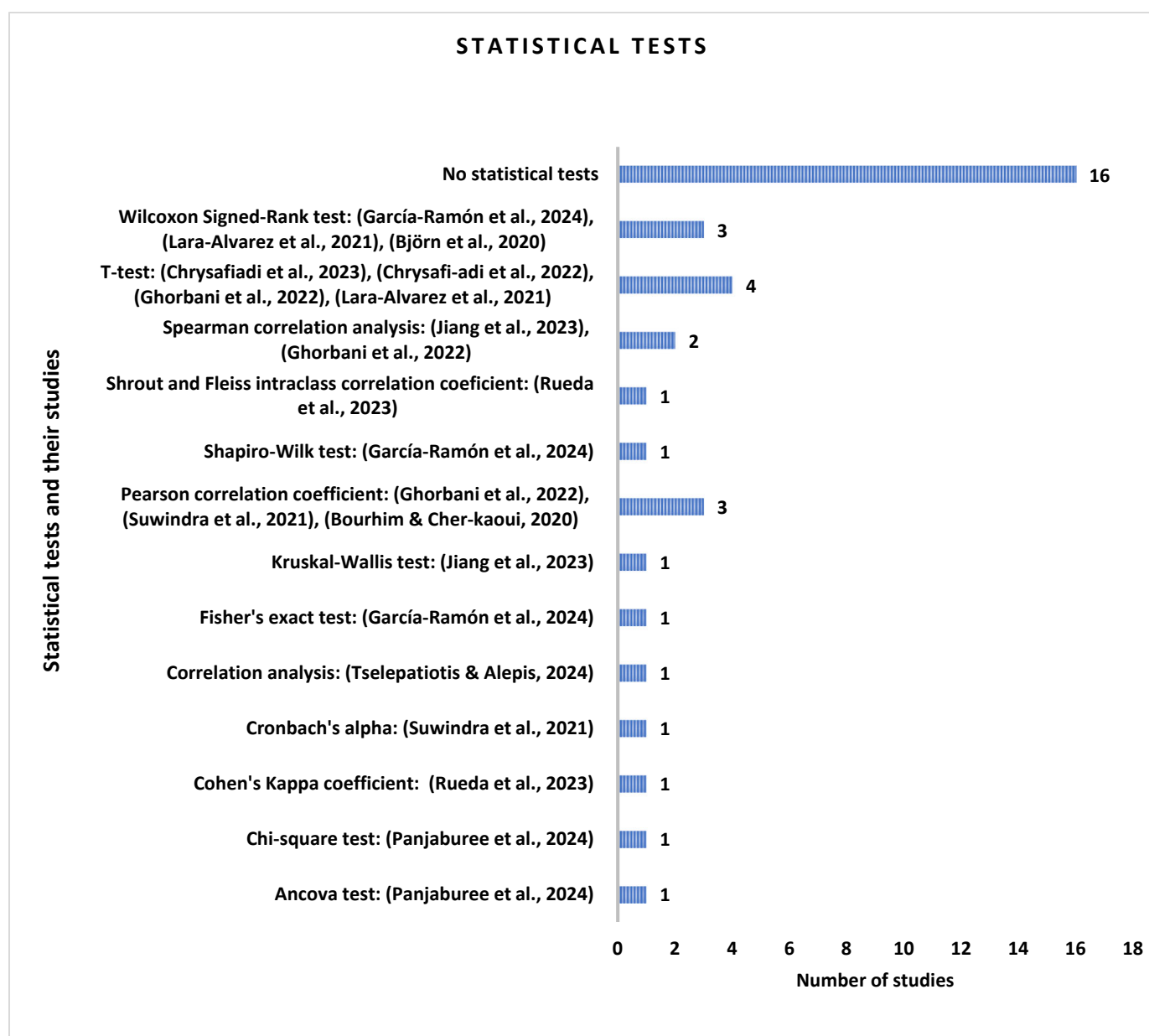


Figure 12. Statistical tests used in the studies [49–51,54,56,57,62,68–70,72,76].

Focusing on the key findings, most studies reported overall positive outcomes related to their proposed approaches in terms of game performance, the effectiveness of the fuzzy logic system implemented in the study, positive player feedback, and improvements in the participants' abilities developed through the game. Further details can be seen in Table 7.

Table 7. Experiments conducted in the studies.

Reference	Year	Participants	Instruments and Metrics	Key Results	Statistical Analysis
[60] (Guniš et al., 2025)	2025	University students (age: 22–24 years) and teachers (age: 30–50 years) generated 64 solutions.	Player’s solution: quantitative and qualitative attributes from each solution (e.g., number of commands, use of recursion, unnecessary commands).	The characterization of solutions with unnecessary commands or applying indirect recursion can be identified via fuzzy attribute implications.	Statistical tests were not used.
[49] (García-Ramón et al., 2024)	2024	53 healthy participants (mean age = 27.88, standard deviation = 9.71) played the tailored game mode (i.e., with dynamic difficulty adjustment) and the non-tailored game mode.	- Electroencephalography (EEG) signals from the participants to compute engagement using two game modes (tailored game mode and non-tailored game mode) and two EEG sensors (Unicorn, Bitalino). - EEG signals were processed using OpenSignals Version 1, EEGLab v2023.1. and Python version 3.10.12. - Questionnaire to collect opinions on the favorite game mode, the easiest game mode to play (ease of play), the least frustrating game mode to play (frustration), the least boring game mode to play (boredom), the game mode with the fastest game response time to commands (game response time), and the most entertaining game mode (entertainment).	Wilcoxon Signed-Rank test reported a significant difference in the engagement index between game modes when the EEG signal was obtained using the Unicorn sensor (p value = 0.04054). Fisher’s exact tests showed significant associations between the game modes (non-tailored, tailored) and the players’ variables: ease of play using Unicorn sensor to collect the EEG (p value = 0.009341), and frustration using the Unicorn sensor to collect the EEG (p value = 0.0466).	- Shapiro–Wilk test. - Wilcoxon signed-rank test. - Fisher’s exact test.
[73] (Liu et al., 2024)	2024	98 students (age 12) participated in the experiments to assess the security knowledge they obtained via the virtual character.	Questionnaire.	Experiments proved that the model can create autonomous virtual characters that help players understand what happens after their actions. The questionnaire results showed that watching how virtual characters react in the game helps players understand safety risks more easily.	Statistical tests were not used.
[75] (Nourian et al., 2024)	2024	Authors organized multiple test-play workshops using a planning scenario: the redevelopment of a factory into an urban neighborhood. They did not indicate the number of participants.	Authors did not mention controlled experiments or surveys to quantify user experience or learning outcomes. The evaluation was qualitative, based on observations, player behavior, and reflections.	The game successfully enabled participatory decision-making. Players were able to reach consensual decisions using the mathematical tools of the game (e.g., opinion pooling, proportional fitting).	Statistical tests were not used.

Table 7. Cont.

Reference	Year	Participants	Instruments and Metrics	Key Results	Statistical Analysis
[62] (Panjaburee et al., 2024)	2024	110 Thai eighth-graders (mean age = 14) participated in the experiments as follows:	- Eye-tracking metrics collected via a Tobii eye tracker and Tobii Pro Lab software were used to analyze players' visual attention (gaze) and engagement. - A questionnaire was used to analyze how players felt about their learning and what motivated them after trying the learning method.	Students who played the personalized game using fuzzy logic and decision trees learned more than those who played the version without personalization.	- ANCOVA test. - Chi-square test.
		57 students participated in the fuzzy logic and decision tree-based personalized gaming approach. 53 students participated in the decision tree-based gaming approach without fuzzy logic and personalized learning manners.		Experimental results reported that the game improved students' digital citizenship outcomes and positively influenced their perceptions.	
				Eye-tracking results indicated that the gaming environment effectively increased student engagement.	
[70] (Tselepatiotis and Alepis, 2024)	2024	23 participants (15 males and 8 females, ages: 17–28).	Questionnaire to collect opinions on tutorial clarity, interest in the game, quality of graphics and sound, attractiveness, and fun.	Most players found the tutorial, goals, and rules of the game clear, giving positive feedback. Players rated the game S enjoyable and educational, indicating that it effectively combined fun with learning.	Correlation analysis.
[56] (Chrysafiadi et al., 2023)	2023	Three groups (ages: 18–20) Total: 102 participants - Group 1 (34 students) played the non-adaptable skill version. - Group 2 (34 students) played the stereotype-based adaptable version of the game. - Group 3 (34 students) played the adaptable skill version of the game.	- Game score. - Questionnaire on ease of play, interest in the game, motivation, and engagement in the learning process.	Results showed that the adapted game (i) reduces negative emotions such as boredom and frustration, (ii) maintains players' interest, and (iii) encourages replayability.	t-test.

Table 7. Cont.

Reference	Year	Participants	Instruments and Metrics	Key Results	Statistical Analysis
[74] (Felix et al., 2023)	2023	52 participants (ages: 13–63 years; mean = 33.55; standard deviation = 12.4; 35 female; 17 male) composed of: 35 professionals from a public health institution, 17 students and independent professionals without any formal employment relationship. Each participant played the game using virtual reality goggles, headphones, and a smartphone.	• Game score. • Questionnaires (pre-test and post-test).	Players reported the highest number of recommendations in the categories of “Gender and Human Rights” (50%) and “Human Rights” (22%), suggesting they mainly struggled with challenges related to violations of women’s basic rights. Results showed that the model adapted its assessment of players’ performance based on how they explored the 360° videos. Consequently, the model helped to identify players with learning difficulties.	Statistical tests were not used.
[50] (Jiang et al., 2023)	2023	• Eight patients with severe hemiplegia (mean age = 57.8 years). • Eight patients with moderate hemiplegia (mean age = 55.2 years). • Eight patients with mild hemiplegia (mean age = 56.7 years). • Eight healthy participants (mean age = 52.7 years).	• Fugl–Meyer Assessment for Upper Extremity (FMA-UE) Scale. • Accuracy from classifiers.	Support vector machines, k-nearest neighbors, regression trees, and fuzzy inference systems (FISs) were compared in terms of accuracy, recall, precision, and F1 score. Evaluations of 32 participants proved that the game-based assessment system effectively differentiated between varying degrees of paralysis, achieving a 93.5% accuracy rate when compared to the therapist’s manual scale.	• Kruskal–Wallis test. • Spearman test
[63] (Jondya et al., 2023)	2023	51 students from middle and high school.	Questionnaire.	82.4% of the participants mentioned that the game’s interactivity and adaptiveness features improved their understanding of historical events related to Indonesian Independence. 80.4% of the participants felt that using the game enriched their experience of learning history.	Statistical tests were not used.

Table 7. Cont.

Reference	Year	Participants	Instruments and Metrics	Key Results	Statistical Analysis
[59] (Krouska et al., 2023)	2023	40 university students (22 males, 18 females, age: 18–22 years old).	Questionnaire for the following aspects: game engagement, experience, educational effectiveness, usability, and user satisfaction.	80% of the students reported high levels of satisfaction with the game engagement, the system’s usability, and the overall user experience. 77.5% of the students reported that the game improved their knowledge and programming skills.	Statistical tests were not used.
[66] (Méndez et al., 2023)	2023	A simulated user.	Output of the fuzzy system.	The fuzzy system can efficiently customize the user interface based on the user’s profile.	Statistical tests were not used.
[69] (Rueda et al., 2023)	2023	36 students who finished the course on electrotherapy.	• Game score. • Assessment from an expert.	Cohen’s Kappa coefficient was applied to evaluate the level of agreement between the expert professor and the serious game on categorical items such as electrode technique, electrode selection, constant current, and constant voltage. The results showed perfect agreement between the expert and the serious game on these items. The Shrout and Fleiss intraclass correlation coefficient was used to assess agreement on continuous items. The results showed moderate to excellent agreement, with moderate agreement specifically for electrode placement, current modulation, and applied intensity.	• Cohen’s Kappa coefficient. • Shrout and Fleiss intraclass correlation coefficient.

Table 7. Cont.

Reference	Year	Participants	Instruments and Metrics	Key Results	Statistical Analysis
[57] (Chrysafiadi et al., 2022)	2022	Experiment 1. Two simulated participants. Experiment 2. 128 students: Group A: 64 students learning the HTML programming language in a course and playing the FuzAd-Escape game. Group B: 64 students learning the HTML programming language in a course only.	Game score. Questionnaire. Test on the HTML programming language.	Experiment 1. The simulations showed that the changes to the game scene were based on the player's knowledge. Experiment 2. A <i>t</i> -test revealed a significant difference between groups (A and B) in terms of the test on HTML programming language. Group A reported better marks on the test than group B.	<i>t</i> -test.
[51] (Ghorbani et al., 2022)	2022	37 participants (16 females, 21 males, age: over 50 years).	- MoCA test on mild cognitive impairments. - Game score.	A high correlation was found between the overall score of the serious game and the overall score of MoCA in the control group, while a moderate correlation was reported in the mild cognitive impairment group.	- Pearson correlation coefficient. - Spearman correlation coefficient. <i>t</i>-test.
[71] (Haryanto et al., 2022)	2022	20 configurations were tested.	Output of the fuzzy system.	The fuzzy system can create a variation in the frequency of item occurrences.	Statistical tests were not used.
[52] (Morales et al., 2022)	2022	20 children from 6 to 12 years old (mean age = 8.3; standard deviation = 1.76) with autism of medium and high functioning levels.	- Number of breaths performed by the children. - Number of prompts needed by the children.	Results showed that the personalization module could adjust the difficulty of the next level based on the player's performance.	Statistical tests were not used.
[64] (Rachmawati et al., 2022)	2022	40 participants.	- Game score. - Assessment of the fuzzy system.	The game "Historical Knowledge Test" showed stable results.	Statistical tests were not used.
[53] (Aziz Hutama et al., 2021)		- Two therapists assessed the game for dysgraphia problems. - Five children to determine whether the game is entertaining.	Questionnaire to assess the children's engagement.	Children reported that the game was fun to play. Experts reported that the game could serve as an alternative medium for therapy.	Statistical tests were not used.

Table 7. Cont.

Reference	Year	Participants	Instruments and Metrics	Key Results	Statistical Analysis
[54] (Lara-Alvarez et al., 2021)	2021	40 participants in secondary school.	• The mean number of stages required to transition a student from an unpleasant emotional state to an alternative emotional condition. • Player’s response time. • Last game score.	Inductive control with the fuzzy logic system improved learning by identifying and encouraging positive emotions.	• <i>t</i>–test. • Wilcoxon signed-rank test.
[65] (Purnamasari et al., 2021)	2021	20 participants.	• Game score. • Questionnaire.	Results showed that fuzzy logic was successful in computing the game score.	Statistical tests were not used
[76] (Suwindra et al., 2021)	2021	18 students (age: 12–14 years).	• Game-factor questionnaire. • Emotional questionnaire. • Character questionnaire. All of them applied in the game.	There was a relationship between game-factor, emotional, and character variables. The neuro-fuzzy approach can identify how game elements, emotional responses, and children’s characteristics are related. This can assist teachers in guiding students to select proper games and manage their emotions, which can influence children’s character development.	• Pearson correlation coefficient. • Cronbach’s alpha.
[68] (Björn et al., 2020)	2020	35 participants (10 males, 25 females, ages: 20–30 years) randomly assigned to three equally sized groups: • Group 1 played the simulator with fuzzy feedback. • Group 2 played the simulator with exact feedback. • Group 3 did not play the simulator.	Questionnaire on the knowledge of the EEG method.	Wilcoxon signed-rank test revealed that the two groups using a computer-based electrode placement simulator reported significant improvement in theoretical knowledge and practical skills compared to the group that learned without the simulator. Results showed that incorporating a simulator into electrode placement training improved students’ practical electrode placement abilities.	Wilcoxon signed-rank test.

Table 7. Cont.

Reference	Year	Participants	Instruments and Metrics	Key Results	Statistical Analysis
[72] (Bourhim and Cherkaoui, 2020)	2020	181 participants (55.8% males, 44.2% females, mean age: 32.25 years).	Questionnaire on virtual reality experience.	77.34% of the participants indicated that it was intuitive to manipulate objects and navigate in the game. 91.39% of the participants reported that the simulation felt realistic. Participants indicated strong satisfaction with the fire simulation system.	Pearson correlation coefficient.
[58] (Chrysafiadi et al., 2020)	2020	Two examples. It is not indicated whether there were real participants or simulated participants.	• Game score • Changes in the player's knowledge level states (novice, average, good, excellent) within the fuzzy system.	The game dynamically adjusted its scenario based on the learners' knowledge level and followed their progress and navigation in the game.	Statistical tests were not used.
[61] (Lee et al., 2020)	2020	74 elementary school students.	Questionnaire.	- Most students provided positive feedback. - The artificial intelligence fuzzy agent for the robotic game of Go was well-received by the participating students.	Statistical tests were not used.
[67] (Ponce et al., 2020)	2020	Not indicated.	Not indicated.	A theoretical model and prototype for a gamified system designed to encourage energy-saving habits in users of smart thermostats.	Statistical tests were not used.
[55] (Robles and Quintero M., 2020)	2020	206 high school students played 5400 games in total.	Game score.	The students improved around 14% in the topics covered.	Statistical tests were not used.

4. Discussion and Future Directions

4.1. Discussion on RQ1—What Is the Taxonomy of Serious Games Using Fuzzy Logic?

Over 80% of the studies implemented educational serious games. An unexpected finding was that only 20% of the studies focused on serious games in the health domain using fuzzy logic. Additionally, all the studies provided visual feedback to the players. It is important to note that 35.71% of the studies included auditory feedback as well. In this regard, a study [80] reported that a key aspect of game accessibility is providing feedback through several channels (visual, auditory, and tactile). Nevertheless, none of the studies provided tactile feedback for players.

Regarding the interaction method, the most widely used interaction methods in the serious games were mouse and keyboard (32%) and smartphone touch screens (32%), as expected. The most widely used platforms were desktops and laptops (50%), followed by smartphones (32%). Although virtual reality has become more affordable and accessible, few studies incorporated it into serious games. Similarly, only a few studies played the game using voice or body movements. A study [81] suggested that game design elements (e.g., platform) should be tailored to the player's profile (e.g., gender, age, culture). For instance, this study reported that people aged 30–35 prefer to play games using consoles, while older adults prefer to play games using tablets and smartphones.

The Entertainment Software Association (ESA) publishes annual statistics on the video game industry in the United States, including aspects such as player perceptions and attitudes, behaviors and preferences, platforms used, motivations for playing, and the top games in the United States. In its 2025 report [82], the ESA presented data for six generations: Gen Alpha (ages 5–12), Gen Z (ages 13–28), Millennials (ages 29–44), Gen X (ages 45–60), Boomers (ages 61–79), and the Silent Generation (ages 80–90). According to the ESA, smartphones were the most widely used platform across all the generations. The second most used platforms were consoles in players aged 5–44 and PC in players aged 45–90.

Based on these findings, future directions for further research in terms of RQ1 could be as follows: (i) to implement serious games focusing on health that use fuzzy logic, so that linguistic variables could be used in the games that are commonly employed in medical environments; (ii) to provide auditory and haptic feedback in game interfaces; (iii) to implement serious games that could be played on various platforms (e.g., console, smartphone, PC, tablet), allowing players to choose their preferred platform; and (iv) to design serious games that allow players to select their preferred interaction method—whether traditional (mouse and keyboard) or alternative (voice, body movements).

4.2. Discussion on RQ2—What Are the Game Design Characteristics?

Regarding the game design, most studies implemented adventure game genres and narratives involving storytelling. However, a study [83] suggested that older adults over the age of 65 prefer casual games (i.e., games with simple interfaces and easy-to-learn game rules). Specifically, this study reported that older people prefer puzzle games, followed by action games. Additionally, player satisfaction is a key aspect of game usability. According to [84], game usability involves four key aspects: (i) learnability—how easy it is to understand the game rules the first time the game is played; (ii) efficiency—the ability of players to perform actions quickly after learning the game; (iii) errors—how easily players can recover from mistakes; and (iv) satisfaction—the degree of approval reported by the player.

Moreover, all the studies implemented games in which the control commands are executed in a fixed way. This reduces usability and accessibility, since players must adapt to predefined interfaces rather than adjusting controls to their own abilities. Based on [80],

adaptive controls (i.e., the ability to remap commands and maximize compatibility with various control devices) should be provided to improve game accessibility. According to [85], game accessibility can be defined as “the ability to play a game even when functioning under limiting conditions. Limiting conditions can be functional limitations, or disabilities—such as blindness, deafness, or mobility limitations”. Evidence from accessibility research supports this. A scoping review [86] reported that flexible input methods, including remappable buttons and controls, and multiple interface options (e.g., joysticks, wearable sensors, and eye gaze tracking), can reduce barriers for players with motor or mobility limitations. These adaptive features allow players to adjust control configurations to reduce fatigue and better meet the needs of different players. Furthermore, the review [86] highlights the importance of integrating accessibility early in game production and indicates that flexible interfaces are essential for making games more inclusive. Similarly, a study [87] identified input remapping for game controls, as well as subtitles for spoken language, adjustable difficulty, and visual clarity, as key features to provide game accessibility for players with functional limitations. Based on these findings, opportunity areas for further research considering game usability and accessibility could be as follows: (i) to provide players with options to select their preferred game genre, which may increase satisfaction and overall usability; and (ii) to design tailored adaptive control interfaces, allowing several options to execute the same command. This would enable players to choose how to interact with the game based on their preferences and needs, thereby reducing inaccessibility.

4.3. Discussion on RQ3—What Is the Aim of Using Fuzzy Logic in Serious Games and How Is the Fuzzy Logic System Implemented?

Most studies (75%: 21 out of 28 studies) implemented fuzzy logic systems in the games involving fuzzification, fuzzy inference, and defuzzification. The remaining studies used only fuzzy sets in the games. Additionally, results showed that the triangular membership function was the most widely used function for converting the crisp inputs into linguistic variables during the fuzzification process (62%: 13 out of 21 studies using fuzzy logic systems). In terms of the fuzzy inference, 90% of the studies (19 out of 21 studies using fuzzy logic systems) used the Mamdani inference method. For defuzzification, the most widely used method to convert the fuzzy output into a crisp value was the center of the area, also known as the centroid method (38%: 8 out of 21 studies using fuzzy logic systems).

Fuzzy logic has mainly been used in serious games to adjust game difficulty by modifying non-player character behavior (38% of the studies: 8 out of 21 studies using fuzzy inference systems).

It is worth noting that none of the studies reported the use of deblurring methods, which can influence visual clarity, immersion, and player comfort in gaming. According to [88], image deblurring is an image restoration process aimed at recovering a clear image from one that has been blurred by motion, defocus, or environmental factors. This review explained several deblurring methods used to restore images to a visually clear state (e.g., Wiener filter, Lucy–Richardson algorithm, convolutional neural networks, recurrent neural networks, and graph convolutional networks).

On the other hand, based on [80], game accessibility can be improved by (i) dynamically adjusting the game difficulty; (ii) maintaining player engagement by modifying the game scene (e.g., providing rewards, adding sensory effects); and (iii) tailoring the game interface to meet the player’s needs—suitable texts and reduced stimuli (e.g., reducing the brightness, adapting the font size, and font type).

Given the results of this study, future directions for further research using fuzzy logic in games could be as follows: (i) to increase player engagement by adjusting game difficulty using fuzzy logic systems that take physiological and behavioral input from the player (i.e., player’s emotions obtained through biosignals—eye gaze, electroencephalography signal,

heart activity, electrodermal activity; player's body movements for serious games focusing on rehabilitation that consider the range-of-motion of the movements based on age and medical condition); (ii) to implement fuzzy logic systems that adapt the game interface to the players' needs (e.g., adjusting the font type and size, modifying audio levels, or changing colors to customize for visual or sensory preferences); and (iii) to integrate fuzzy logic with deblurring algorithms to improve gaming experiences.

4.4. Discussion on RQ4—Were Experiments Conducted in the Studies?

Approximately 79% of the studies included human participants in their experiments. Additionally, only 23% of these studies used small sample sizes, with fewer than 30 volunteers. Regarding the instruments and metrics, the most widely used instruments and metrics to assess the games were questionnaires (58%: 15 out of 26 studies), followed by game scores (38%: 10 out of 26 studies). It is important to note that only a few studies used physiological data to evaluate the games (i.e., player engagement computed via electroencephalography signals [49], player's visual attention computed via an eye tracker [62], and player's breathing [52]). Finally, it is worth noting that over half of the studies (57%: 16 out of 28 studies) did not conduct any statistical analysis on their results.

Based on these findings, opportunity areas in terms of RQ4 could be as follows: (i) to use instruments and metrics in the experiments that analyze physiological data; for instance, bio-signals (e.g., eye gaze, electroencephalography signal, heart activity, electrodermal activity) can provide more accurate insights into players' emotional responses than relying only on questionnaires; and (ii) to conduct statistical analyses on the results using appropriate tests (e.g., correlation tests to determine associations between study variables, parametric or non-parametric tests to determine whether there is a significant difference in a study variable between study groups), so that the findings of the studies could be more reliable and ensure that results are not due to random chance.

4.5. Risk of Bias of the Studies

The JBI's critical appraisal tool for Quasi-Experimental studies [89] was employed to assess the risk of bias of the 28 studies included in this review. Specifically, this tool consists of nine questions that analyze bias related to temporal precedence, to selection and allocation, to confounding factors, to administration of intervention/exposure, to assessment, detection, and measurement of the outcome, to participant retention, and to statistical conclusion validity. Each question can be answered using "Yes", "No", "Unclear", and "N/A" (Not applicable). "Yes" was selected when the criterion was explicitly addressed and clearly met in the study. "No" was chosen when the information was available and showed that the criterion was not met, and "Unclear" was selected when the study did not provide enough information to make a judgment. All the studies described the cause-and-effect relationship (question 1: bias related to temporal precedence). Only a few studies (17.86%: 5 out of 28 studies [50,51,57,62,68]) included a control group in the experiments (question 2: bias related to selection and allocation). Regarding bias related to confounding factors (question 3), just over a quarter of the studies (28.57%: 8 out of 28 studies [50,52,55–57,68,73,74]) reported that participants in the comparisons were similar. For bias related to administration of intervention/exposure (question 4), almost half of the studies (42.85%: 12 out of 28 studies [49–52,56,57,59,63,69,70,73,74]) indicated that participants, apart from the intervention of interest, received similar conditions. With respect to bias related to assessment, detection, and measurement of the outcome, few studies (14.28%: 4 out of 28 studies [50,55,68,74]) conducted pre- and post-intervention measurements (question 5), most studies (60.71%: 17 out of 28 studies [49–57,59,63,68–70,72,74,76]) indicated that the outcomes of the participants were measured in the same way across the groups (question 6),

and most studies (71.43%: 20 out of 28 studies [49–57,59,62,63,65,66,68–70,72,74,76]) explained the instruments used to measure the outcome reliably (question 7). In terms of bias related to participant retention, half of the studies (14 out of 28 studies [49,50,52–54,57,59,62,65,68–70,72,76]) indicated that the follow-up was completed or provided explanations for group differences due to participant dropout (question 8). Finally, almost half of the studies (42.85%: 12 out of 28 studies [49–51,54,56,57,62,68–70,72,76]) employed appropriate statistical analyses (question 9: statistical conclusion validity). Details on all nine questions of the tool were provided in only two studies [50,68]. Details on the results of this tool for each study are presented in Supplementary Material.

4.6. Limitations

This scoping review presents the following limitations:

Single-reviewer bias. Only the author (EJRR) screened and assessed the articles; consequently, this might have introduced a risk of bias in the scoping review.

Limitations associated to a scoping review. For instance, protocols for analyzing article quality (e.g., Qualsyst [90], quality assessment with diverse studies (QuADS) [91]) used in systematic reviews were not applied. Therefore, there could be articles with low rates in terms of these protocols.

Keyword limitations. The search terms were selected based on the most widely used and recognized terminology in the reviews presented in Table 1. Several keywords could be associated with ‘serious games.’ Based on [18–20,22–28], ‘game’ and ‘serious game’ were included as search terms. Nevertheless, it is important to consider that studies using related terms such as ‘video games’, ‘gamification’, ‘digital games’, ‘computer games’, ‘technology’, ‘human-computer interaction’, or ‘application’ may have been excluded. Similarly, based on [16,27], ‘fuzzy logic’ was used as a search term to include fuzzy systems; consequently, studies referring to ‘fuzzy theory’, ‘fuzzy sets’, ‘neuro-fuzzy’, or ‘ANFIS’ (Adaptive Neuro-Fuzzy Inference System) may have been excluded as well.

Database access limitations. Although nine databases were consulted, not all the articles retrieved were accessible. Consequently, articles indexed in other databases or inaccessible were not considered.

5. Conclusions

This scoping review analyzed the state-of-the-art in serious games that use fuzzy logic. A total of 494 articles were retrieved through searches conducted in nine databases (ACM Digital Library, IEEE Xplore, IOPscience, MDPI, PubMed, ScienceDirect, Springer, Wiley, and Web of Science). Of these, only 28 articles met the inclusion criteria. This study addressed four research questions. Regarding “RQ1—What is the taxonomy of serious games using fuzzy logic (i.e., application area, activity, modality, interaction method, environment, and hardware architecture)?”, it can be concluded that most of the serious games focused on education (80% of the studies), while the remaining studies addressed health applications. The majority of the studies used serious games in which the participants performed mental activity (89.28% of the studies). All games provided visual feedback, and only 37.21% of the studies included auditory feedback as well. The most widely used interaction methods were mouse and keyboard (32% of the studies) and smartphone touch screen (32% of the studies). Over half of the studies used 3D games (61%). Desktops and laptops were the most used platforms (50% of the studies) followed by smartphones (32% of the studies). Focusing on the “RQ2—What are the game design characteristics (i.e., game genre, narrative, and game rules—how is it played)?”, the results of this review showed that most of the studies (35.71%) implemented the adventure genre, including different storytelling elements (e.g., exploring a water village and an island with castles, understanding

scenarios related to digital citizenship, and historical events), followed by the simulation genre, which involved several scenarios (e.g., urban planning, clinical situations, disaster mitigation) (33.33%). Regarding how the games were played, over half of the studies (64%) required participants to choose options or objects presented in the game interface. In terms of “RQ3—What is the aim of using fuzzy logic in serious games and how is the fuzzy logic system implemented (i.e., fuzzification, fuzzy inference, defuzzification)?”, this review found that most studies (75%) implemented fuzzy logic systems that involved fuzzification, fuzzy inference, and defuzzification processes. The triangular membership function was the most widely used method for fuzzifying data (62%: 13 out of 21 studies using fuzzy logic systems). Mamdani inference was employed in 90% of the studies (19 out of 21 studies using fuzzy logic systems), while the center of area method was the most widely applied technique for defuzzification (38%: 8 out of 21 studies using fuzzy logic systems). Fuzzy logic was applied in serious games for three key purposes: (i) adjusting game difficulty (38% of the studies: 8 out of 21 studies using fuzzy inference systems); (ii) providing tailored feedback within the game (19%: 4 out of 21 studies using fuzzy inference systems); and (iii) calculating game scores (14%: 3 out of 21 studies using fuzzy inference systems). Finally, with respect to “RQ4—Were experiments conducted in the studies (i.e., participants, instruments and metrics, key results, and statistical analysis)?”, this review identified that a total of 79% of the studies involved human participants in their experiments. Most of the studies employed instruments and metrics to assess their proposals (93%: 26 out of 28 studies). Questionnaires (58%: 15 out of 26 studies) and game scores (38%: 10 out of 26 studies) were the most widely used instruments and metrics in the studies. Nevertheless, over half of the studies (57%: 16 out of 28 studies) did not conduct any statistical analysis of their findings.

Several opportunity areas have been identified and discussed for each research question. Future work should focus on developing serious games that provide multimodal feedback—visual, auditory, and haptic channels—to players; designing tailored adaptive control interfaces that offer several choices to execute the same command; and implementing fuzzy logic systems that dynamically adjust game interfaces and game difficulty based on players’ needs and players’ physiological signals.

Supplementary Materials: The following supporting information can be downloaded at: <https://www.mdpi.com/article/10.3390/technologies13100448/s1>, Table S1: JBI tool.

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